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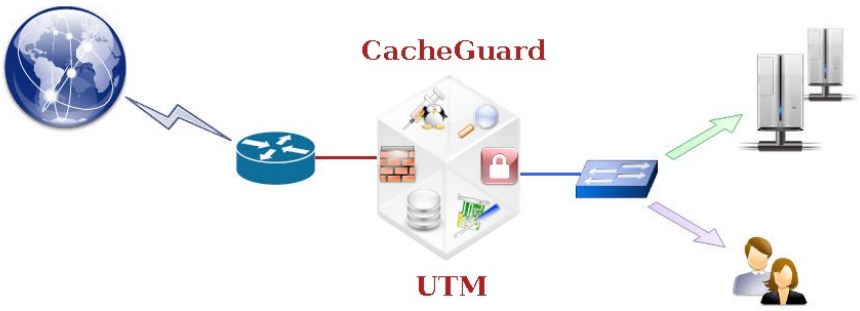
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CacheGuard Overview

The User's Guide enables you to quickly and clearly learn how to configure and administer a CacheGuard appliance. For a detailed description of each command used in this guide, please refer to the Commands Manual. The web administration GUI provided with CacheGuard-OS acts as a front end to the CLI and is therefore not documented separately.

CacheGuard Gateway allows you to connect your networks to the internet securely while protecting your IT infrastructure against harmful traffic. It also enables you to provide the QoS (Quality of Service) required by critical network traffic such as VoIP. What makes CacheGuard Gateway a unique solution is its dual functionality: in forwarding mode, it protects users accessing the internet, while in reverse mode, it protects web applications. To deploy a CacheGuard Gateway, simply install CacheGuard-OS on the machine of your choice. The only requirement is that the machine must have at least two NICs (Network Interface Cards). For assistance with CacheGuard-OS installation, please refer to the Getting Started section.



CacheGuard-OS integrates a range of network security and traffic optimisation features including, but not limited to, a firewall, VPN, web antivirus, filtering proxy, reverse proxy, WAF, traffic shaping, and web caching. All these features can be securely and efficiently activated simultaneously on the same machine, ensuring optimal use of the underlying hardware resources.

Using CacheGuard-OS

Implementing and configuring CacheGuard-OS is straightforward and efficient, even if you are not a networking or security specialist. With CacheGuard-OS, the complexity of the integrated open-source components is hidden beneath the surface, allowing you to simply operate a single system and benefit from a powerful and reliable engine.

CacheGuard Gateway Functions

CacheGuard-OS Network Optimisation

- Web caching
- Web traffic compression
- Web application load balancing
- QoS & Traffic shaping
- Caching DNS

CacheGuard-OS IP Security

- Internal, external, and auxiliary zoning
- Forward and reverse web proxy
- Transparent HTTP proxy
- Proxy chaining and parallel deployment
- Access lists
- IP firewall with NAT and PAT
- IPsec VPN in site-to-site or remote access modes
- Protection against SYN flood, port scanning, spoofing, and more

CacheGuard-OS Web Security

- URL guarding based on URL blacklists, whitelists, and regular expressions
- URL guarding policies based on access time, IP, and LDAP requests
- Automatic blacklist updates
- Web Application Firewall (XSS, SQL injection, etc.)
- Access logging
- LDAP and Kerberos AD authentication
- SSL termination

CacheGuard-OS High Availability

- RAID capabilities
- Backup and restore on a spare machine
- Ethernet link bonding
- VRRP redundancy
- Multi-WAN support

- SSL mediation/inspection
- Web gateway antivirus
- Antivirus service for email

CacheGuard-OS Web Optimisation

- Persistent web caching
- HTTP compression
- Web cache sharing
- Traffic shaping

CacheGuard-OS Administration

- CLI (Command Line Interface) configuration
- Console port administration
- Remote administration via Web GUI and SSH
- Logging to remote syslog servers
- SNMP agent and trap generation

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Change Logs & Release Notes

Version UF-2.6.2 (29 September 2026)

- The DHCP range configuration bug introduced in version UF-2.6.1 has been fixed.
- The HTTP Compression mode has been fixed to do not close the transfer with outstanding read data remaining and to compress data in case where the caching is enabled.
- The appliance now supports host-routed /32 (255.255.255.255 network mask) configurations on its external network interface with an off-link default gateway.
- DHCP and PPPoE support have been added for the external network interface.
- The bug in the Administration Web GUI of an appliance installed as a Manager that prevented its IP address from being modified has been fixed.
- Minor bug fixes have been applied.

Version UF-2.6.1 (7 September 2026)

- The arbitrary command injection vulnerability that could be exploited during configuration loading by an administrator has been fixed.
- The **admin ssh key** ... usage form of the **admin** command has been improved to allow an SSH public key to be set inline.
- The **guard** and **transparent** modes have been jointly enhanced to transparently filter unwanted domain names based on the SNI field of HTTPS traffic.
- The **qos shape router** ... usage form of the **qos** command has been improved to allow port range specifications.
- The DHCP server has been enhanced to also assign dynamic IP addresses to devices connected to the **auxiliary** interface. Therefore, the **dhcp** command syntax has been modified to specify the network interface when defining dynamic IP ranges and fixed IP addresses.
- The syntax of the **authenticate** command for LDAP servers has been modified. It is no longer necessary to specify an IP address for an LDAP server in addition to its hostname.
- The **firewall** command has been fixed to display the destination port specification when no NAT specifications are present.
- Routed traffic can now be shaped for distinct functional interfaces (associated with 802.1Q pseudo-interfaces). Therefore, the syntax of the **qos shape router** command usage form has been changed; the **internal** interface can no longer be used. If the OS is upgraded using a patch, all QoS router rules associated with the **internal** interface are automatically translated to the **web** interface.
- QoS has been fixed to properly shape routed traffic through the **external** logical interface when SNAT mode is activated.
- QoS has been fixed to classify next-peer traffic as web traffic via the external interface. In addition, the **peer** command has been enhanced to allow a QoS percentage value to be specified for each next peer.
- The listening ports for ISAKMP and NAT-T services used by the IPsec VPN server can now be set to low-numbered ports (less than or equal to 1024).
- The implementation of next and previous peer configuration has been revised to use httppeer ports (see the **port** command). If you are using chained peers, your configuration should be reviewed.
- The **access** command now supports the **access fweb** ... usage form to restrict access to the explicit web proxy based on the *X-Forwarded-For* header. Consequently, the **access web** ... usage form now defines only the networks allowed to access the explicit web proxy directly.
- The network specifications for explicit web clients, transparent web clients, and previous peers are now independent. Refer to the **access**, **transparent**, and **peer** command documentation for further information.
- The forwarding proxy configuration has been enhanced to allow antivirus and compression modes to be enabled simultaneously.
- The URL list loading policy has been improved. An attempt to update a URL list that is more than 30 days out of date now automatically triggers a full reload in create mode.
- The pull and push gateway configuration on the Manager has been fixed to properly handle the client certificate control file. As a result, template creation based on a gateway configuration now correctly excludes client certificates from the certificate control file.
- On the Manager, configuring a gateway or template based on another gateway or template configuration (configuration inheritance) now also includes client certificates.
- The OS has been fixed to prevent the removal of unused temporary working directories after a certain period on the Manager, which could make it impossible to manage configurations.
- The memory usage estimation during OS installation and tuning has been revised.
- The boot issue occurring after the installation of the 32-bit OS on machines with 3 GB of RAM or less has been fixed.
- The entire documentation (User's Guide and Command Line Interface documentation) has been reviewed.
- The CacheGuard-OS License Agreement has been upgraded to version 2.9.
- Squid is no longer used as a reverse proxy, even in non-WAF and non-sticky connection load-balancing modes.
- The Auditing Web GUI has been fixed to properly display guarding logs.
- The reverse proxy now supports HTTP/2.

- The Administration and Auditing Web GUIs have been enhanced to comply with Content Security Policy (CSP) requirements, with JavaScript interactions modernised using HTML5 data attributes.
- The embedded BeVyPN application has been enhanced to comply with Content Security Policy (CSP) requirements, with JavaScript interactions modernised using HTML5 data attributes.
- The appearance and usability of the Web GUI have been reviewed and improved.
- The Kerberos initialization bug has been fixed.
- The OS now supports both MIT and AD™ (Active Directory) style Kerberos authentication. Therefore, the syntax of the **authenticate ad** command usage form has been modified, and the **authenticate hapassword** command usage form has been removed.
- The **authenticate kerberos encrypt** command usage form has been removed, allowing the OS and the Kerberos server to negotiate the highest available encryption type.
- Squid has been upgraded to version 7.6.
- Apache Web Server has been upgraded to version 2.4.67.
- ClamAV has been upgraded to version 1.5.2.
- StrongSwan has been upgraded to version 6.0.6.
- ModSecurity has been upgraded to version 2.9.13.
- OWASP CoreRuleSet has been upgraded to version 3.3.9.
- jQuery has been upgraded to version 3.7.1.
- Minor bug fixes have been applied.

Version UF-2.5.2 (10 April 2026)

- OS upgrading on the Manager has been corrected to ensure that the OS version is also updated on managed gateways.
- Connections and disconnections to the IPsec VPN server are now logged in a separate log file called **vpnipsec** and are no longer included in the system logs.
- The Windows PowerShell IPsec VPN access configuration has been corrected to call the setupVPN function with @() instead of @('') when split tunnelling is not used (see the **vpnipsec access conf windows**... command).
- The Windows PowerShell IPsec VPN access configuration has been corrected to properly delete existing certificates before installing new ones (see the **vpnipsec access conf windows** command).
- Logs can now be sent to syslog servers using TCP or UDP over SSL (the affected command is the **log** command).
- Gateway unavailability detection in static IP routes has been improved.
- In a routing table where different pinged servers are specified for the same gateway, gateway unavailability detection has been fixed.
- Minor bug fixes have been applied.

Version UF-2.5.1 (20 February 2026)

- The **system report load** command has been fixed to display the correct CPU and I/O load values, expressed as relative percentages over time.
- The **system report cpu** command has been added to the system to display CPU usage as a percentage of total capacity.
- The **vpnipsec access conf**... command has been corrected to generate valid local identifiers, thereby resolving the bug introduced in the previous version.
- Dynamic DNS updating capability has been incorporated into the OS. The new **dynamicdns** command enables the configuration of the Dynamic DNS service in use.
- Minor bug fixes have been applied.

Version UF-2.4.2 (5 February 2026)

- OpenSSL® has been upgraded to its latest stable version (3.6.0).
- OpenSSH® has been upgraded to its latest stable version (10.2).
- OpenLDAP® has been upgraded to its latest stable version (2.6.10).
- Apache® web Server has been upgraded to its latest stable version (2.4.66).
- Squid® has been upgraded to its latest version (7.3).
- StrongSwan® has been upgraded to its latest version 6.0.4).
- CURL® has been upgraded to its latest stable version (8.18.0).
- FTP-based file transfers have been corrected to ensure that TLS is enforced during the authentication phase when the target FTP server supports TLS authentication.
- The use of DSA keys with SSH has been permanently disabled.
- The **apply** command has been corrected so that it no longer erroneously reports that the appliance does not have sufficient RAM to activate the new configuration in certain cases.
- The **admin ssh fingerprint** command on the Manager has been corrected to display the contextual RSA key fingerprint.
- In the WAF, the processing order has been changed so that custom WAF rules are evaluated before WAF filters.
- Minor bug fixes have been applied.

Version UF-2.4.1 (5 January 2026)

- The licensing model has been revised to make the use of CacheGuard-OS completely free, regardless of the number of users supported by the Gateway or the number of Gateways managed by the Manager.
- The **vpnipsec access conf android** *client-tls-id* **show** command has been enhanced to generate a VPN profile that includes a proxy section.
- The **vpnipsec access conf apple** *client-tls-id* **show** command has been fixed to generate a VPN profile that

uses the currently configured proxy port, as set by the **port proxy** command, instead of the fixed value of 8080.
- Minor bug fixes have been applied.

Version UF-2.3.6 (2 December 2025)

- The bug introduced in the previous version that prevented the deployment of the generated system CA certificate has been fixed.
- The import (loading) of previously imported third-party CA certificates has been fixed to function correctly.
- Integrity checking has been fixed so that configurations containing HTTPS reverse websites referring to a non-existent intermediate certificate can no longer be applied.
- Minor bug fixes have been applied.

Version UF-2.3.5 (18 October 2025)

- The ClamAV® antivirus has been upgraded to its latest stable version (1.4.3).
- The antivirus updater script has been improved to prevent it from running more than once per hour.
- The bug introduced in the previous version that prevented OS installation on a Microsoft® Hyper-V® virtual machine has been fixed.
- The entire documentation (User's Guide and Command Line Interface documentation) has been reviewed.
- Minor bug fixes have been applied.

Version UF-2.3.4 (2 October 2025)

- The Linux® kernel has been upgraded to the version 6.6.100.
- The Manager bug that was preventing to properly pull a Gateway configuration after a certain period of time since its registration on the Manager, has been fixed.
- A 2FA (2 Factor Authentication) mode for remote administrator logins (SSH and Web Administration GUI) has been added to the OS.
- The OS behaviour has been modified to do not allow the Auditing Web GUI activation in case where the Administration Web GUI is deactivated.
- The embedded BeVyPN application has been improved to support 2FA (Two Factor Authentication).
- Minor bug fixes have been applied.

Version UF-2.3.3 (12 June 2025)

- The Administration Web GUI has been fixed to hide the new OS availability icon after a successful OS upgrade to the latest version.
- The **vpnipsec access conf windows** command has been fixed to correctly generate the PowerShell® script in case where the IPsec VPN server in access mode is in a split tunnelling mode (only some selected local networks are configured to be accessible via the IPsec VPN).
- Maximum values for score thresholds in the WAF have been increased to 25 (instead of 9). This makes it possible to detect false positive matches in a staging environment with the help of the Auditing Web GUI.
- ModSecurity® has been upgraded to its latest version (2.9.8).
- OWASP® CoreRuleSet has been upgraded to its latest stable version (3.3.7).
- The **conf factoryreset** command bug generating a 104 error (input/output issue) has been fixed. This bug is introduced when upgrading the OS version from UF-2.1.3 to UF-2.2.1 by applying a patch.
- The bug in the **rweb** command that let the logical configuration in an inconsistent state after the deletion of a unique host has been fixed.
- Minor bug fixes have been applied.

Version UF-2.3.2 (25 March 2025)

- The **cache report** command has been fixed to display the correct web cache report.
- The installation program has been fixed to exclude USB disks from the list of available disks when installing the OS from a USB stick.
- The software defined network has been modified to always activate the path checker even in a single gateway mode.
- The configuration file upload via the Administration Web GUI has been improved to support large files and all kinds of web browsers.
- Minor bug fixes have been applied.

Version UF-2.3.1 (13 March 2025)

- The OS has been upgraded to support the 255.255.255.254 network mask (prefix 31).
- The guard filter deletion has been fixed to properly remove the deleted filter from guarding policies.
- A new usage form has been added to the **vpnipsec** command to generate IPsec VPN profile (or script) file to use on remote client systems when the IPsec VPN is configured in access mode.
- The system can now send some information (such as IPsec VPN client configurations) by email. To this end, the **email** command has been enhanced to configure the email server and credentials to use to send emails.
- The keyword **wait** has been added to commands that normally run in background to specify that the command should be executed in blocking mode (e.g. **apply** command).
- The **file** command has been enriched to allow to delete a file on a remote file server.
- Minor bug fixes have been applied.

Version UF-2.2.2 (31 July 2024)

- OpenSSH® has been upgraded to its latest stable version (9.8p1).
- Apache® web Server has been upgraded to its latest stable version (2.4.62).
- Squid® has been upgraded to its latest version (6.10).
- StrongSwan® has been upgraded to its latest version (5.9.14).
- Duplicated lease end time errors and records in the DHCP report have been fixed.
- The IPsec VPN report has been fixed to properly display certificate based authenticated connections.
- The Kerberos® initialisation issue has been fixed.
- Minor bug fixes have been applied.

Version UF-2.2.1 (6 April 2024)

- The Linux® kernel has been upgraded to the version 6.6.14 and all drivers have been upgraded to support the latest hardware in the market.
- The installation program has been fixed to allow an installation from a USB memory stick.
- The firewall has been improved to support the SIP protocol.
- The bug which prevented the restore operation since the UF-2.0.1 OS version has been fixed.
- The reverse web mode has been enhanced to allow configurations in which backend web servers (real hosts) are accessed via the external network interface or via site to site IPsec VPN tunnels established with the system. In addition, the reverse web proxy can now communicates with real hosts. As a consequence, the syntax of the **rweb** command has been changed (see the **rweb host** usage form). In case where the appliance is upgraded using a patch, the **rweb** interface and the **http** protocol are used for existing configurations.
- The installation program has been enhanced to support an installation on a machine with only 512 MB of RAM.
- The size of the installation CDROM image has been reduced.
- The PXE installation program has been enhanced to support UEFI based machines (64 bits only).
- Minor bug fixes have been applied.

Version UF-2.1.3 (25 January 2024)

- The web access module has been fixed to allow clients that are connected via the 802.1Q pseudo-interfaces (in VLAN mode) to use the embedded web proxy.
- The **firewall** command has been improved to allow the modification of default limits for DoS (Denial of Service) attack. See the **firewall dos** command for further information.
- The default maximum number of TCP new connections (SYN) per source IP address has been raised. New default values can be obtained using the **firewall dos** command.
- The access command has been improved to allow the specification of 802.1Q pseudo-interfaces in web access rules (see the **access web** command).
- In absence of any other specifications, a blank manager template is now initialised with a default system CA certificate and a default server certificate.
- Minor bug fixes have been applied.

Version UF-2.1.2 (11 December 2023)

- The proxy, the name server and the antivirus have been upgraded to their latest stable versions.
- The IPsec VPN server has been enhanced to allow remote VPN servers having a non fixed IP address to establish a site to site IPsec VPN tunnel.
- Minor bug fixes have been applied.

Version UF-2.1.1 (1 December 2023)

- The OS has been adapted to suit Microsoft® Azure® and Amazon® AWS® clouds requirements.
- The installation program has been improved to detect NVMe® and virtual block based disks.
- Default associations between physical and logical network interfaces have been changed on a gateway system. Now by default, the external interface is associated to eth0 and the internal interface is associated to eth1.
- The generated system CA certificate during the first appliance start-up, is now properly installed.
- Admin access management has been fixed to properly allow newly added IPs.
- The **password** command has been enhanced with the possibility to modify both the console and Web Administration passwords in a single operation.
- The CDROM installation program has been enhanced to support UEFI based machines (64 bits only).
- The Linux® kernel has been upgraded to the version 4.19.288.
- The ssh password authentication can now be disabled (see the **admin ssh password** command).
- The bug that was blocking explicit log rotations in case where the **web** (or **tweb**) mode is deactivated, has been fixed.
- Minor bug fixes have been applied.

Version UF-2.0.2 (11 April 2023)

- The **waf** command has been fixed to do not erase the bypass application set for a website in case where the bypass rule list is erased for that website.
- The WAF Auditing Web GUI has been fixed to properly decode HTML encoded data in POST requests.
- The **system report service** command has been fixed to display the DNS server state.
- Now the DNS server can be queried even if the **web** and **tweb** are both disabled.

- The appliance access manager has been fixed to take into consideration override names setup with the **ip name** command. In addition, in case of any modification in override names, the Firewall and QoS are restarted.
- The **rweb site del**... command has been fixed to do not remove back end web servers associated to a website in case where the deleted website name will remain present for another protocol (http or https). The fix has been also applied to other configuration related to reverse websites (**rweb via**..., **waf rweb**...).
- The Administration Web GUI automatic logout has been fixed.
- Now the **apply check** command displays possible warnings.
- The routing issue via the auxiliary network interface has been fixed.
- The firewall rules management module has been fixed to do not apply the default policy to new connections incoming from the internal zone in case where the VLAN mode is deactivated and rule sets other than the **web** rule set are not empty.
- The firewall has been fixed to allow fragmented IP packets in IPsec traffic.
- The SSL mediation mode has been fixed to automatically download intermediate certificates even in case where the web access has been restricted by the **access web**command.
- The **tls** command has been enhanced to allow for setting the size of generated RSA private keys.
- The bug that was preventing to install the OS without the web caching feature has been fixed.
- Minor bug fixes have been applied.

Version UF-2.0.1 (3 February 2023)

- A new command called **manager** have been added to the system to manage remote gateways. To use this command, the OS should be installed as a *manager* system (as opposed to a system installed as a *gateway* system).
- The Linux® kernel has been upgraded to the version 4.19.231.
- All open source packages have been upgraded to their latest versions and have been rebuild from scratch using latest GNU C library (glibc-2.35) and GNU C compiler (gcc-11.2.0).
- The new usage form **system report connection** has been added to the **system** command to display the number of active connections with the appliance.
- The new usage form **system report antivirus** has been added to the **system** command to display the status of the last automatic antivirus signatures update.
- The apply reporting has been fixed to report 100% (instead of 99%) when the antivirus update is fully completed.
- Trusted CA certificates have been updated from Mozilla as of: Thu Sep 30 21:39:27 2021 GMT. The OS has been enhanced to automatically update trusted CA certificates once a month.
- The system has been upgraded to support TLS 1.3.
- The maximum period for log retentions has been modified from 365 to 366 days (available during installation only).
- Some basic open source software have been upgrade to their latest stable versions.
- The **tls** command has been enhanced to allow the generation of certificates that do not use OCSP.
- The **tls server** and **tls ca** command usage forms of the **tls** command have been changed to be uniform with the **tls client** usage form.
- The syntax of **tls** command has been changed. Now to manage server certificate the **server** keyword should be systematically used as the first argument. To manage the system CA components (certificate and key) the **system** keyword should be specified after the **ca** keyword. To manage third-party CA certificate the **third** keyword should be specified after the **ca** keyword. To import third-party CA certificate, the **load** keyword replaces the **import** keyword.
- The **apply** command has been modified to automatically generate new TLS objects in case where explicit TLS generations or loadings are note invoked.
- Restricted administrator users can now be deleted properly without generating an error during the apply operation.
- The syntax of the **admin** command has been changed for SSH key management. Now an identifier should be associated to an SSH key first. Then, its content can be loaded from a trusted file server.
- Now, the first time a new restricted administrator is logged in, she/he is invited to modify her/his password.
- The **cancel** command can now be invoked by restricted administrator users without generating any errors.
- OWASP® Core Rule Set (CRS) has been upgraded to its latest version (3.3.2). This involves the renaming and renumbering of generic filters.
- CAUTION: the syntax of the **waf** has been modified as follows: "waf rweb bypass" becomes "waf rweb bypass **rule**". Also generic WAF filters has been renamed and renumbered. If your configuration includes the bypass of some generic rules in order to avoid false positive matches, you are invited to review your configuartion. Please refer to the documentation for further details.
- Now the blocking of web requests/responses by the WAF is based on an anomaly scoring principal. Please refer to the documentation for further details.
- The WAF module has been enhanced to offer the following functionality: blocking of DoS (Denial of Service) attacks, blocking of requests based on IP reputation, blocking of requests coming from a particular country, bypass of generic filters based on the type of the application (WordPress®, Drupal®...). Please refer to the documentation for further details.
- The logging has been enhanced to allow the activation or deactivation of logging on remote syslog servers as per the type of traffic (see the command **log**).
- The syntax of the **authenticate**... command has been slightly changed. Please refer to the documentation for details.
- The **ip**, **access** and **vpnipsec** commands have been improved to accept IP addresses in CIDR notation (in the form *<ip/prefix>* instead of *<ip>* *<netmask>*).
- The limitation associated to the IPsec VPN usage in a multi WAN configuration has been removed. Now it is possible to route IPsec traffic via a master gateway and automatically switch the routing via a backup gateway in case of a failure on the master gateway. See the **vpnipsec** command manual for further information.

- A new command called **file** has been added to the system to load or save all files related to the configuration in a single operation.
- The rights of restricted administrator users have been changed. Now restricted administrator users can only read (consult) the system configuration. Restricted administrator users are now called unprivileged administrator users (refer to the **admin** command for further information).
- The **traceroute** command has been added to the system.
- The association of multiple client SSL certificates to the SVMP-v3 user name has been removed from the system (now only one client SSL certificate can be associated to the SVMP-v3 user name). See the **admin** command for further information.
- The management of TLS chain certificates has been modified. Now when defining an HTTPS reverse website, you have the possibility to specify an intermediate CA certificates. Please refer to the **tls** and **rweb** commands for further information.
- The system integrity checking has been modified to allow the deactivation of **web**, **tweb** and **rweb** modes at the same time.
- The CacheGuard-OS License Agreement has been upgraded to version 2.5 to include OS installation as a *manager* system.
- Lots of minor bugs have been fixed.

Version EH-1.5.5 (20 April 2021)

- The QoS controller has been fixed to properly shape the traffic on the external and auxiliary interfaces when the VLAN mode is activated.
- The system has been improved to avoid any latency in web browsing during the antivirus update process. This requires about 1280 KB of additional RAM so a RAM upgrade may be needed on the target machine.
- Minor bug fixes have been applied.

Version EH-1.5.4 (16 March 2021)

- The issue that was slowing down the AV signatures downloads has been fixed.
- TCP communications have been tuned to get better performances.
- The proxy, the VPN and the antivirus have been upgraded to their latest stable versions.
- Minor bug fixes have been applied.

Version EH-1.5.3 (11 February 2021)

- The installation program has been fixed to include all required SCSI drivers in the boot loader initial RAM disk.
- The access control module has been fixed to properly allow administration accesses when the VLAN mode is activated.
- The installation module has been fixed to detect VirtIO devices.
- In order to comply with the RFC 5280, the "OCSP Signing" Extended Key Usage has been removed from generated X509v3 certificates (only the "TLS Web Server Authentication" Extended Key Usage is kept).
- Minor bug fixes have been applied.

Version EH-1.5.2 (2 December 2020)

- The network name resolution module has been enhanced to resolve the embedded OCSP responder host name to the system's external IP address (external VRRP IP in HA mode). This helps to avoid asymmetric routing when the **ocsp** mode is activated.
- Minor bug fixes have been applied.

Version EH-1.5.1 (31 October 2020)

- The IPsec VPN support has been added to the system and a new command called **vpnipsec** has been added to manage IPsec VPNs. Both site to site (**site**) and remote **access** VPN configurations are supported. The embedded forwarding web proxy and resources behind the embedded firewall can securely communicate via the IPsec VPN.
- The Linux® kernel has been upgraded to the version 4.9.230 and all drivers have been upgraded to support the latest hardware in the market.
- The reporting in the Administration Web GUI dashboard has been fixed to properly refresh all reports (including reports on NICs and disks).
- The **system report cpu** usage form of the **system** command has been replaced by **system report load**.
- The **tls** command has been enhanced to allow the loading of a CSR file in order to generate signed certificate by the system's CA certificate. With this enhancement the system can now act as a mini PKI.
- When a CA certificate is added to the system it is automatically considered as a trusted CA for web browsing. Now it is possible to do not trust a CA certificate for browsing by specifying the optional **off** argument when adding the CA certificate with the **tls** command. In this case the CA certificate can only be for other purposes (such as the VPN server).
- The **authenticate ldap certificate** usage form of the **authenticate** command has been removed. If an LDAPS server SSL certificate has to be verified against a CA certificate, the CA certificate should be imported first using the **tls** command and then the CA certificate verification can be activated using the **authenticate ldaps ca ...** command. In case where the system is upgraded using a patch, an existing LDAPS CA certificate is purged and then it should be configured again manually.
- The authentication type for SNMP-v3 user has been changed from SHA-1 to SHA-256.
- The **md5** and **sha** (for SHA-1) authentication hash functions are no longer allowed for SNMP-v3 traps. Allowed authentication hash functions are now: **sha256**, **sha384** and **sha512**.

- Access policies to (from) the appliance from (to) remote networks/hosts have been reinforced by the specification of the involved network interface. Therefore, the syntax of the **access** command has been changed. In case where the system is upgraded using a patch, an access entry for every interface is added to the system and access policies should probably be reviewed after having patched the system.
- When defining a transparent network with the **transparent** command, the network interface from which traffic are transparently caught should be specified now. In case where the system is upgraded using a patch, the same transparent network is added for every interface and transparent networks should probably be reviewed after having patched the system.
- Minor bug fixes have been applied.

Version EH-1.4.2 (29 June 2020)

- The ICAP service not restarted with the previous patch is restarted to properly handle the brotli compression format.
- The **tls** command has been enhanced to allow you to create client certificates signed by the system's CA certificate. Client certificates can be used to authenticate VPN clients (VPN features are coming soon).
- The **tls ca** command can now be used to add and import an intermediate CA as well as a root CA.
- Self signed SAN certificates generation has been fixed to properly generate a self signed certificate and not a CA certificate.
- Certificates can now be revoked with the **tls** command.
- An OCSP (Online Certificate Status Protocol) responder has been added to the system. You can use the **tls** and **port** commands to configure it. Use the **mode** command to activate it.
- Minor bug fixes have been applied.

Version EH-1.4.1 (11 June 2020)

- A new mode called **tnat** (for transparent NAT) has been added to the system. When the **tnat** mode is deactivated, web traffic in transparent mode go to the internet with their real IP addresses (and are not source NAT with the appliance's external IP address).
- The transparent mode activation has been moved from the "[GENERAL]/[Main Settings]/[Main Features]" page to the "[NETWORK]/[Main Settings]/[Network Services]" page in the Administration Web GUI.
- The transparent command has been enhanced to take into account the QoS for traffic exchanged via the auxiliary interface.
- The DNS has been fixed to properly listen on VRRP IP addresses.
- The proxy configuration has been fixed to add the X-Forwarded-For header to all HTTP(S) requests if at lease one next peer is configured.
- The maximum period for log retentions has been modified from 31 to 365 days (available during installation only).
- Internal access policies have been modified to allow the connect method to ports 1024-49151 (in addition to the port 443) from the forwarding proxy
- The CSR and signed certificate generation programs has been fixed to properly handle Certificate Signing Requests and CA signed certificates for SAN certificates.
- Failed login via the Administration Web GUI are now logged and reported with SNMP traps and syslog alerts.
- Some additional ciphers has been added to the SSH server. This command prints a report on the current running operation in background. The Administration Web GUI has also been enhanced with an animated icon to show the current running operation.
- A new command called **job** has been added to the system.
- Now the antivirus uses HTTPS instead of HTTP to download virus signatures.
- Reports displayed in the Administration Web GUI dashboard are now automatically updated.
- The option **report** has been added to the **qos** to print a report on the traffic managed by the QoS controller.
- Minor bug fixes have been applied.

Version EH-1.3.7 (29 May 2018)

- The CacheGuard-OS License Agreement has been upgraded to version 2.4.
- The bug making the main proxy to crash while the web mode is deactivated has been fixed.
- The **system end** command has improved to display the *scheduled* state when the subscription renewal is scheduled for the next day.
- A new command named **keyboard** has been added to the system. This command allows you to set the console key map.
- The installation module has been improved to allow the creation of partitions larger than 2TB.
- The SNMP agent has been enhanced to give SSDs lifetime.
- The **ip** command has been fixed to do not allow names for pinged servers in static routes.
- Minor bug fixes have been applied.

Version EH-1.3.6 (9 March 2018)

- The web proxy default access rights have been modified to allow the "PATCH" method in both forwarding and reverse modes.
- The **waf** command has been enhanced to offer the possibility to globally allow or deny the "PATCH" method as an insecure HTTP method.
- Custom WAF rules has been extended to support the "PATCH" HTTP method.
- The **apply** command has been fixed to properly check the integrity of destination NAT rules and do not erroneously produce the error 212.

- The internal firewall rules have been fixed to properly allow DHCP request broadcasts and lease renewals.
- The **dhcp report** command has been fixed to display DHCP lease end times in local time instead of UTC time.
- Minor bug fixes have been applied.

Version EH-1.3.5 (8 January 2018)

- The antivirus module has been enhanced to bypass a white list of domain names. Therefore the **antivirus whitelist** usage form of the **antivirus** command has been modified to allow you to define a white list of domain names as well as a white list of virus signatures.
- The web proxy default access rights have been modified to allow the "PUT", "DELETE" and "TRACE" methods in both forwarding and reverse modes.
- The **setup** command has been enhanced to use dialogues boxes.
- The dialogue box version of the **setup** command has been enhanced to allow you to set the timezone in the virtual edition.
- The virtual edition has been enhanced to set the console keyboard layout during the first startup.
- The bug making the guarding module to crash with some malformed URLs has been fixed.
- The textual configuration view in the Administration Web GUI has been improved to have a more user-friendly representation of the whole configuration.
- Minor bug fixes have been applied.

Version EH-1.3.4 (17 September 2017)

- Communication between the appliance and the patch download service has been switched from HTTP to HTTPS.
- Communication between the appliance and the subscription/registration service has been enhanced to support HTTPS.
- Custom WAF rules have been enhanced to allow the specification of more than one HTTP method separated by the pipe character.
- The **waf** command has been enhanced to offer the possibility to globally allow or deny insecure HTTP methods such as "PUT", "DELETE", "CONNECT" and "TRACE".
- Minor bug fixes have been applied.

Version EH-1.3.3 (4 September 2017)

- The bug that was preventing big objects caching has been fixed.
- ICP (RFC 2187) has been replaced by HTCP (RFC 2756) for communications between cache peers.
- The **dns** command has been enhanced to allow the explicit resolution of all names to IPs.
- Custom WAF rules has been extended to support the "PUT", "DELETE", "CONNECT", "OPTIONS" and "TRACE" HTTP methods.
- Minor bug fixes have been applied.

Version EH-1.3.2 (20 July 2017)

- The Web cache is no longer cleared after the antivirus activation.
- The RAM vs HDD capacities tuning has been improved to have better performance for the caching.
- The memory consumption for the caching has been improved and the usage of the available persistent cache has been reviewed accordingly.
- Due to the instability of the compress mode while combined with the antivirus mode, the compress mode is automatically disabled for forwarding web traffic when the antivirus mode is activated. This fix should be considered as a workaround before the complete resolution of the issue in future releases.
- The bug that was preventing to save logs has been fixed.
- The bug that was preventing to activate the DHCP server has been fixed.
- Due to a high number of InvalidState and IllegalSyn rejected TCP packets on networks, rejected InvalidState and IllegalSyn TCP packets are no longer logged.
- Minor bug fixes have been applied.

Version EH-1.3.1 (29 June 2017)

- The antivirus module can now be used as a service by external systems such as an MTA (Mail Transfer Agent).
- The antivirus module has been enhanced with the possibility to integrate a white list of virus names to eliminate false positive matches.
- The authentication mode has been upgraded to support the Kerberos® protocol.
- The **system report** usage form of the **system** command has been enhanced to print the total number of blocked or allowed contents.
- The WAF has been enhanced to offer the possibility to expose original HTTP error messages generated by backend web servers.
- The time format in all logs has been changed to be compliant with the RFC3339 (with the caveat that the time offset format may not be respected for some logs).
- The compression module has been fixed to properly compress javascript files.
- The embedded firewall has been enhanced to protect against UDP flood attacks.
- The CA certificate bundle has been updated to its latest version.
- The subscription system has been fixed so the renewal of an expired subscription takes into account the date of purchase as the start date. In this case, the renewal is done for the given period rounded to the nearest whole day.
- The subscription system has been fixed so the reactivation of a suspended appliance is completed without errors.

- The dashboard layout has been enhanced.
- A **Donate** button has been added to the Administration Web GUI of the free edition in order help us to maintain CacheGuard-OS and and develop new features
- The trial period has been extended from 15 to 21 days.
- Minor bug fixes have been applied.

Version NG-1.2.6 (29 December 2016)

- The upgrade from v1.2.4 to v1.2.5 by applying a patch has the side effect that some processes swap on disk. This issue has been fixed by applying a patch to upgrade to the present version. Note that for the v1.2.5, CacheGuard-OS requires a minimum of 1 GB of RAM to activate the antivirus mode. Therefore if the antivirus mode is activated on an appliance running under the v1.2.5, the applying of the patch may require upgrading the RAM on the target machine. Otherwise the appliance may stop working properly.

Version NG-1.2.5 (21 December 2016)

- A workaround has been added to the system to resolve the inability of some Microsoft® OS's to download updates while the compress mode is activated.
- The upload of a local configuration file from the Administration Web GUI has been fixed to support web browsers other than Firefox®.
- The Administration Web GUI has been upgraded to support Microsoft® IE11®.
- The antivirus basic module has been upgraded to its latest version.
- Minor bug fixes have been applied.

Version NG-1.2.4 (20 October 2016)

- The firewall and access control modules have been fixed to support the active FTP protocol with a data port other than 20 (in EPRT and PORT mode).
- The forwarding proxy has been modified to allow the HTTP method OPTIONS. However the method OPTIONS remains denied for reverse websites.
- The CacheGuard logo has been slightly modified.
- The **conf** command has been fixed to properly save authentication modes.
- Network diagrams in the User's Guide have been enhanced.
- The dashboard in the Administration Web GUI has been enhanced to display the available OS updates and the end of the system subscription.
- Minor bug fixes have been applied.

Version NG-1.2.3 (11 September 2016)

- The free edition has been limited to 5 users in forwarding mode and 3 users in reverse mode.
- The syntax of the **register** command has been modified according to the new licensing terms.
- The SSL mediation has been modified to allow the usage of 3DES algorithm to encrypt data between the system and target HTTPS servers.
- The **authenticate mode** usage form of the **authenticate**command has been changed.
- The automatic loading of a URL list has been improved so in the case where a URL list has never been loaded, it is entirely loaded from scratch.
- The bug in the Administration Web GUI that prevented adding new SNMP traps has been fixed.
- The installation program has been fixed to properly generate the default and CA certificates.

Version NG-1.2.2 (2 June 2016)

- A critical bug fix related to the basic forwarding proxy module has been integrated into the system. The bug made the appliance totally unstable.
- The bug that was preventing to start the integrated DHCP server has been fixed.
- Minor bug fixes have been applied.

Version NG-1.2.1 (2 May 2016)

- The **guard** command has been enhanced with the possibility to update an existing rule without changing its order in the guard list.
- The CLI has beed enhanced with the possibility to move an element in an ordered list. Commands in question are: **ip**, **guard**, **qos** and **firewall**.
- The Administration Web GUI has been fixed to not change the order of a guard rule in the guard rule list when its associated URL lists are updated.
- The Administration Web GUI has been fixed to not erase the list of URL lists associated to a guard rule when the order of that rule is modified in the guard rule list.
- The redirection to an error page has been fixed in the guarding module to work properly in conjunction with the SSL mediation module.
- Minor bug fixes have been applied.

Version NG-1.2.0 (6 April 2016)

- An SSL mediation (sometimes called inspection) mode has been added to the system. This mode allows you to

cache HTTPS traffic and/or block unwanted contents in HTTPS traffic. The **mode** command has been updated to allow you to activate this new feature (**mode sslmediate on**) and the new command **sslmediate** has been added to the system in order to configure the SSL mediation module.
- A new command named **urllist** has been added to the system. This command replaces the **guard category** usage form of the **guard** command. URL lists can be used by the **guard** command but also the new command **sslmediate**.
- The IP routing has been enhanced to support the usage of multiple gateways to route the traffic to the same network.
- The Linux® kernel has been upgraded to the latest stable version.
- Concurrent accesses to loaded files have been improved to avoid any file overwriting.
- The system patching has been improved to load patches directly from official CacheGuard servers.
- A download progress bar has been added to backup management and patching pages in the Administration Web GUI.
- The installation program has been enhanced to detect USB Ethernet adapters.
- All major basic modules have been upgraded to their latest versions.
- The **access** command documentation has been fixed (removal of **rweb** access).
- Some network activity reporting has been added to the system.
- A dashboard has been added to the Administration Web GUI.
- The udpeer and tcpeer ports have been respectively renamed to icppeer and httppeer.
- The SFTP is now supported to load/save files.
- Some default port numbers have been changed.
- Some minor errors have been fixed in the documentation.
- Minor bug fixes have been applied.

Version NG-1.1.5 (3 December 2015)

- The value of the "X-Forwarded-Proto" header which is added to requests sent to backend web servers (in reverse mode) has been fixed as follows: the value "http" or "https" is set depending on whether the client used HTTP or HTTPS to connect to cloaked web servers.
- The **system** command has been enhanced to display the CPU architecture (32-bit or 64-bit).
- The patching system has been fixed in order to avoid the applying of the same patch more than once.
- The patching system has been fixed in order to create new empty directories.
- The guard management module has been fixed to update guard rules when a guard policy is deleted.
- The bug in the HA module that blocks the AH protocol used to authenticate HA nodes has been fixed.
- The firewall module has been fixed to not block IGMP snooping when the HA mode is activated.
- The bug that was producing erroneous ARP announcements in HA mode has been fixed.
- Internal firewall rules have been reinforced.
- The logging of denied IP packets has been enhanced to report information about the rejection reason.
- The antivirus basic module has been upgraded to its latest version.
- The firewall basic module has been upgraded to its latest version.
- Some minor enhancements have been done.
- Minor bug fixes have been applied.

Version NG-1.1.4 (10 October 2015)

- The bug introduced in version 1.1.3 that prevented to automatically update guard categories has been fixed.
- The security of the VRRP has been enhanced.

Version NG-1.1.3 (5 October 2015)

- The reverse web mode has been enhanced to allow the specification of a port number and QoS for backend web servers. After having applied a patch the default port and QoS will respectively be 80 and 100. Therefore the syntax of the **rweb** command has been modified for the usage form **rweb host**.
- The usage form **access rweb** of the **access** command has been suppressed.
- The reverse web load balancing has has been enhanced to allow the specification of a session cookie generated by web applications running on backend web servers. Therefore the syntax of the **rweb** command has been modified for the usage form **rweb balancer**.
- The bug making the guarding policy inconsistent when one of its guard filters has been deleted has been fixed.
- The bug making the configuration of patched system inconsistent after a factory reset has been fixed.
- The HA basic module has been upgraded to its latest version.
- The authentication module has been expanded with a test option.
- Minor bug fixes have been applied.

Version NG-1.1.2 (2 September 2015)

- The CacheGuard OS License Agreement has been upgraded to version 2.1
- IPV6 has been disabled in the Linux® kernel.
- The issue to access via the HTTPS proxy to the https://outlook.office365.com website (and similar websites that preferably use IPV6 IP addresses) has been resolved.
- The LDAP authentication module has been optimised so all communications with LDAP servers are forced to use IPV4 only.
- The authentication module has been enhanced to allow LDAP binding during the basic authentication phase instead of comparing the entered password to a predefined password attribute.
- The authentication module has been fixed to allow distinguished names containing white spaces. In the case

where the OS is upgraded using a patch the authentication LDAP request should be redefined (see the **authenticate ldap request** command).
- Minor bug fixes have been applied.

Version NG-1.1.1 (1 August 2015)

- The firewall has been fixed to properly manage other protocols than TCP and UDP.
- The IPv6 has been added to the list of supported protocols by the firewall.
- The TLS component management module has been optimised in order to avoid restarting some services when it's useless.
- The bug making a custom WAF rule inconsistent when it contains a star has been fixed.
- Several minor bugs have been fixed in the Administration Web GUI.

Version NG-1.1.0 (13 July 2015)

- **Note**: Note that to upgrade from version NG-1.0.15 to version NG-1.1.0 you should first apply a patch to upgrade to the version NG-1.0.16. Therefore you would be able to upgrade from version NG-1.0.16 to version NG-1.1.0. Patche files are available at www.cacheguard.net/cacheguard-patch.html.
- The installation program has been fixed to report warnings in respect to setup configurations.
- The reverse mode has been enhanced for HTTPS websites to add an "X-Forwarded-Proto http" header to HTTP requests sent to backend web servers (useful for some known applications)
- The **apply** command manual has been completed to give additional information in respect to errors reported during the process of checking the RAM capacity.
- A RAM upgrade is now automatically applied after a reboot.
- The **system** command has been enhanced to check for new updates.
- The bug in the **conf** command which caused the saving of wrong values for the QoS attached to "tweb internal" queue been fixed.
- All commands that use a network name parameter (such as access or rweb) have been enhanced to check if the given name is a FDN (Full Distinguished Name).
- The syntax of WAF rules defined in a flat file has been changed. The keyword **regexp** has been replaced by **uri** and **body**. A new feature has been added to WAF rules to allow filtering based on source IP addresses. The keyword **ip** holds this position.
- The DHCP server has been modified to configure DHCP clients with a web proxy based on the proxy PAC file (ha.pac) delivered by the system.
- A new feature has been added to the WAF to allow you to bypass false positive matches.
- OWASP® rule set for the WAF has been upgraded to its latest version.
- The CacheGuard logo has been modified.
- Minor bug fixes have been applied.

Version NG-1.0.16 (13 July 2015)

- **Note**: Note that no OS has been released for this version but only patch files. Patch files are available at www.cacheguard.net/cacheguard-patch.html.
- The patching program has been fixed in order to properly patch the configuration DB.

Version NG-1.0.15 (9 May 2015)

- The SNI (Server Name indication) support has been added to generated SSL certificates. Therefore more of the same IP address can be shared by multiple HTTPS websites.
- The TLS/SSL support has been hardened to ensure a higher security level.
- Minor bug fixes have been applied.

Version NG-1.0.14 (28 January 2015)

- The bug that was causing incorrect dimensioning of the antivirus capacity has been fixed. To fix this issue you should reinstall the appliance as there is no available patch to address this issue (unless you have a support contract).
- Minor bug fixes have been applied.

Version NG-1.0.13 (22 January 2015)

- The high availability management module has been enhanced to not change the state (**failover** or **active**) of a system in HA mode after an apply operation.
- The installation program has been enhanced to report paying configurations.
- The **conf** command has been fixed to properly manage the transparent port configuration (**port thttp**).
- Minor bug fixes have been applied.

Version NG-1.0.12 (29 December 2014)

- The bug making the guards auto update to crash in case of a communication problem with a file server has been fixed.
- The antivirus basic module has been upgraded to its latest version.
- The automatic logout problem in Administration Web GUI has been fixed.

- Minor bug fixes have been applied.

Version NG-1.0.11 (24 November 2014)

- The bug introduced in version 1.0.10 that prevented having more than one reverse website associated to the same IP address has been fixed.
- The bug introduced in version 1.0.10 that prevented activation of the web server when the reverse website list contains HTTPS websites has been fixed.
- Generic WAF rules associated to a reverse website are reset when a reverse website is deleted.
- The **rweb** mode has been enhanced to redirect HTTP to HTTPS for HTTPS reverse websites (if the IP address associated to HTTPS website is used for an HTTP website).
- The bug that was making the reverse website list unsorted after deleting and adding a new website has been fixed.

Version NG-1.0.10 (17 November 2014)

- The reverse website module has been enhanced to deny attempts to access to website names that are not explicitly defined in the system.
- The configuration saving module has been fixed to properly save all generic WAF rules.
- The bug preventing the main proxy to start when the transparent mode is deactivated has been fixed.
- SSH keys management has been improved.
- Some minor display issues have been fixed in the Administration Web GUI.

Version NG-1.0.9 (31 October 2014)

- The apply operation has been improved to ensure that in a High Availability configuration, master IP addresses are owned by an appliance once all functional services have been started on that appliance.
- The Administration Web GUI has been fixed to properly insert, add and remove elements in lists in the same submitted operation (for firewall rules for instance).
- The Administration Web GUI has been fixed to allow you to enter a six digit value for the maximum cached object size (*cache-maxobject.apl* page).
- The bug that was preventing the activation of the firewall in the following two conditions has been fixed: a non empty auxiliary firewall rule set and the auxiliary network interface not bound to a physical NIC.
- The bug producing an "illegal instruction" error in some virtualization systems has been fixed.
- The console port attached to a serial port is no longer activated if the target machine doesn't have a serial port during the installation.
- The web proxy accessibility has been modified to allow web traffic incoming from all network interface devices but the external interface.
- The transparent feature has been changed to transparently catch web traffic incoming from all network interface devices but the external interface. In previous versions, only web traffic incoming from the internal interface (**web** interface in **vlan** mode) were caught.
- The configuration saving process has been fixed to properly save new guard auto update configurations.
- The traffic shaper module has been improved to allow web traffic shaping exchanged with the auxiliary network interface.
- Administration Web GUI pages to save or load the configuration have been regrouped in the same page and improved to allow saving/uploading the configuration to/from the local machine.
- The bug in the *gui/transparent.apl* page that blocks post and reload content operations has been fixed.
- The Administration Web GUI for the page *gui/qos-shape-gateway.apl* has been enhanced with tabs.
- In the Administration Web GUI, font sizes have been reduced and all icons in the top bar have been grouped to the left.
- The caching system has been enhanced to allow the caching of objects greater than the configured max object size for a limited part of the persistent cache.
- The syntax of the **cache** command has been changed to configure a lower and upper size limit for cached objects.
- A new MIB definition has been released for the SNMP agent to include the size of the reserved area on the persistent cache for very big objects.
- The bug that was preventing the OS installation on a Microsoft® Hyper-V® VM has been fixed.
- The bug that was dropping default traffic in case of an empty shaping rule set for routed traffic has been fixed.
- The QoS rule compilation for reverse websites has been optimised.
- The bash shellshock vulnerability has been fixed.
- Some other minor bugs have been fixed.

Version NG-1.0.8 (24 September 2014)

- The Auditing Web GUI has been fixed to properly display post arguments without evaluating html tags.
- The cache size and memory usage tuning has been reviewed according to recent statistics.
- The installation program has been enhanced to allow the deactivation of some features that require lots of storage space on disk. This allows you to install the system on machines with low storage capacity.
- The bug related to the premature display of the termination message of some system operations has been fixed.

Version NG-1.0.7 (11 September 2014)

- The Administration Web GUI has been improved to allow the displaying of long list in different pages.
- The bug that was preventing to apply custom WAF rules has been fixed.

- The configuration saving has been modified to save restricted administrator users. The saving is limited to login names only (passwords and configurations related to each restricted administrator are not saved).
- In order to ensure command syntax coherency the keyword **raz** has been replaced by **clear** in the **waf rweb custom** command usage form.
- Some other minor improvements have been added to the system to manage the configuration.
- Some other minor bugs have been fixed.

Version NG-1.0.6 (1 September 2014)

- The High Availability module has been improved to ensure that master IP addresses are owned by an appliance once all functional services have been started on that appliance.
- The email syntax checking has been fixed to allow the usage of dash in an email address.
- The reporting of the antivirus last update date has been fixed to display the actual date.
- The configuration settings present in the form of a list have been modified in two ways: lists are kept sorted if the order of elements in the list is not significant. TLS objects and guard categories are some examples of those lists. For ordered lists like firewall rules and shaping rules for routed traffic, the keyword **insert:** has been added to the related management commands to allow the insertion of and element before another one. This avoids to have to save, edit and load the configuration settings.
- The Administration Web GUI has been improved to allow the insertion of elements in lists subject to insertion. The look and feel of pages managing those list has also been improved.
- The **firewall** command has been change to allow the definition of rules with **any** as the output network interface.
- In order to avoid a false positive rule matching by the configured WAF for the Administration Web GUI, **tftp** has been renamed to **ftp_trivial** in the **firewall** command.

Version NG-1.0.5 (7 August 2014)

- A major bug preventing the main proxy to start in some circumstances has been fixed. These circumstances were as follows: transparent mode is off, the transparent list contains at list one network and at least one share or HA peer is configured.
- Backup and restore operations have been fixed to restore properly restricted administrators.
- The patching operation has been optimised but remains manual until a possible next version.
- A new command named **ha** (High Availability) has been added to the system. Combined with the argument **master**, this command allows you to make an attempt to reactivate a master appliance which has been marked faulty without needing to reboot the appliance.
- The Administration Web GUI has been improved to log administrators logins.
- The panel board menu has been improved to allow access to menu items without sub menus.

Version NG-1.0.4 (29 July 2014)

- A critical bug regarding the activation of VLANs and bonding interfaces making the appliance inconsistent and out of service has been fixed (bug introduced in error in the precedent version).
- The timezone setting during the installation has been taken again into account (bug introduced in error in the precedent version).
- The firewall configuration has been modified to give higher priority to NAT rules than the SNAT (Source NAT) mode.
- The IP routing configuration has been fixed so route to a single host can operate properly.
- The installation program has been enhanced to set a default value for the WAF limit files which is less than or equal to the maximum size for uploaded file given during the installation.
- The Administration Web GUI bug that was preventing the activation of administration services has been fixed.
- IP routing has been fixed to allow routing via a gateway connected to the auxiliary interface.
- The bug that was preventing to have an IP route without having a default route has been fixed.
- The textual configuration displaying has been enhanced.
- The auto logout for the Administration Web GUI has been fixed.

Version NG-1.0.3 (21 July 2014)

- The Administration Web GUI bug that was preventing to change the Administration Web GUI password with a password containing the characters '$' or '!' has been fixed.
- The verification of input values for online commands and the Administration Web GUI has been enhanced.
- The firewall module has been fixed to allow traffic with **any** as the protocol.
- The web authentication module has been fixed to allow authentication for reverse websites.
- The Administration Web GUI has been improved to avoid false positive matches by generic WAF rules.
- The FTP proxy over HTTP has been fixed to display properly directory contents.
- The bug that was preventing to have more than one HTTP reverse websites configured with the same IP address has been fixed.
- Some other minor bugs have been fixed.

Version NG-1.0.2 (14 July 2014)

- The bug that was preventing HTTPS reverse websites deleting has been fixed.
- The apply operation integrity check has been fixed to prevent error messages when the VLAN mode is activated.
- When a reverse website present in several forms (HTTP, HTTPS, with multiple IP addresses) is deleted, all its configurations (backend web servers, load balancing, standby mode...) are preserved until the deletion of its last occurrence.

- The total arguments length limit for the Administration Web GUI has been fixed so the firewall can be configured properly using the Administration Web GUI.

Version NG-1.0.1 (12 July 2014)

- A minor bug related to the synflood tuning has been fixed
- A minor bug related to the URL blacklist message page has been fixed.

Version NG-1.0.0 (CacheGuard OS Version 6) (30 June 2014)

- The QoS management has been enhanced and the syntax of the **qos** command has also been changed. Now the keyword **bandwidth** should be specified to define bandwidth limits and the borrowing of the excess bandwidth can be activated or deactivated using the keyword **borrow**. Also the traffic shaping could be specified as a percentage or as a fixed value in Kbps. Other traffic than the traffic destined to the appliance itself (the gateway) could also be shaped in this version.
- Keywords "intern" and "extern" have been respectively renamed to **internal** and **external**.
- The CacheGuard License Agreement has been upgraded to version 2.0. In this version all CacheGuard Software components are subject to the GNU General Public License v3 while the aggregation of those components and other Open Source Software (as OSI definition) forming the "CacheGuard OS" is licensed under the "CacheGuard OS License Agreement version 2.0".
- An SNMP agent and trap sender has been added to the system to monitor the appliance. Please use the **admin snmp** command to configure the SNMP monitoring.
- Keywords related to SSL/TLS have been changed in the **admin** and **authenticate** commands for coherency purposes.
- A new feature has been added to the system so the system can be backed up and restored (see the **system backup** command)
- A contact email address may be specified with the **register** command.
- The Linux® kernel has been upgraded to a latest stable version.
- The access list management for file and monitoring servers has been improved to allow the adding of host names in addition to host IPs.
- The ntp server management has been improved to allow the adding of ntp server names in addition to ntp server IPs.
- The keyword **https** in the **admin** command has been renamed to **tls** as the generated certificate is used by both the Administration Web GUI and the SNMP agent.
- A new logical network interface named **auxiliary** has been added to the system. You can use it for your specific needs (for instance to implement a DMZ or a Back Office zone).
- The syntax of the **firewall** command has been changed.
- Software RAID support has been added to the system.
- The OS is now available in two versions: 32-bit and 64-bit.
- New filter types based on time, authentication and IP ranges have been added to URL guarding module. Therefore the syntax of the command has been completely reviewed. Please refer to the **guard** command documentation for further information.
- In the **authenticate** command, the argument "attribute" has been renamed to **request**.
- Transparent traffic is clearly distinguished from forwarding traffic. A dedicated port is used for the transparent mode and a new command named **transparent** can manage the transparency for selected networks. Also the QoS module manages forwarding and transparent traffic separately.
- The syntax of the **rweb** command has been changed so the management of SAN and wildcard certificates became easier.
- The load balancing policy may be configured for reverse websites. This new feature includes the possibility to have sticky connections.
- The installation program has been improved.
- In order to ensure command syntax coherency the "access" and "virus" logs have been renamed respectively to to **web** and the **antivirus**.
- The logging module has been enhanced to allow the logging of denied packets by the IP firewall.
- In logging mode (mode log is activated) each log type (firewall, web, rweb, guard, virus, waf) can selectively be activated or deactivated.
- Many minor bugs have been fixed.

Version 5.7.7 (4 July 2013)

- The **system** command has been improved to display the subscription end date.
- A new feature has been added to the health panel to report the status of the antivirus and URL guard updates.
- The Administration Web GUI look and feel has been enhanced.
- Minor bug fixes have been applied.

Version 5.7.6 (28 February 2012)

- A new subscription verification module has been added to the system.

Version 5.7.5 (20 October 2011)

- Several minor bugs have been fixed in the Administration Web GUI.
- The log rotation system has been fixed to properly rotate the WAF log.

Version 5.7.4 (12 October 2011)

- The OS has been improved to support a better crash recovery.

Version 5.7.3 (15 August 2011)

- Installation in test mode has been improved to allow choosing the OS to load at bootup.
- The mgt (for management) vlan has been renamed to mon (for monitoring).
- The keyword mgt (for management) in the **access** command has been renamed to mon (for monitoring).

Version 5.7.2 (11 July 2011)

- The command "factoryreset" has been removed and replaced by the argument **factoryreset** added to the **conf** command.
- Reporting capabilities has been added to the proxy cache module (see the **cache report** command).
- Reporting and health checking capabilities has been added to the system (see the **system report** command).

Version 5.7.1 (20 June 2011)

- The proxy cache module has been upgraded to its latest version.
- The bug that was preventing to download small video files while the cache and antivirus mode are both enabled has been fixed.
- The command "filter" was renamed to **waf** (for Web Application Firewall).
- In the **mode** command, the keyword "filter" was renamed to **waf**.
- In the command "log", the keyword "filter" was renamed to **waf**.
- In the **antivirus** command, the keyword "clear" was renamed to **create**.

Version 5.6.10 (27 May 2011)

- The antivirus no longer checks images and textual contents.

Version 5.6.9 (20 May 2011)

- The minor bug related to the PUA mode activation has been fixed in the Administration Web GUI.
- All diagrams in the documentation has been enhanced with new icons.

Version 5.6.8 (16 May 2011)

- The persistent caching module has been enhanced for better disk caching performance.
- The antivirus module has been upgraded to its latest version to fix several internal bugs.
- The antivirus no longer checks video contents.
- The syntax of the **antivirus maxobject** command has been changed.
- Now files larger than the limit configured with the **antivirus maxobject** command won't be scanned by the antivirus.
- A new command named **setup** has been added to the system. This command is automatically executed when you first connect to the system.

Version 5.6.7 (25 April 2011)

- The Administration Web GUI look and feel has been enhanced.
- The Auditing Web GUI has been fixed to display properly all virus and guard logs.
- Now the **antivirus update report** command also displays the last automatic AV update.

Version 5.6.6 (20 April 2011)

- The minor bug related to the trial version initial date has been fixed.

Version 5.6.5 (14 April 2011)

- Now the proxy is allowed to connect to ports between 1024 and 49151.
- The bug that was preventing clients connecting to the internal DNS when the appliance doesn't use itself as a DNS has been fixed.

Version 5.6.4 (4 April 2011)

- The CacheGuard Logo has been changed.
- The licensing key system has been revised.
- The Administration Web GUI look has been revised.

Version 5.6.3 (2 February 2011)

- The integrated AntiMalware software has been upgraded to its highest version.

- The Auditing Web GUI has been improved to show denied URLs and attempts to access Malware.
- Accessing websites that use NTLM / SSPI authentication works now with latest IIS web servers when the compress or filter mode are activated.

Version 5.6.2 (28 January 2011)

- Setting auto update for blacklists has been fixed in the Administration Web GUI.
- The whole documentation has been reviewed.
- Some other minor bugs have been fixed.

Version 5.6.1 (17 June 2010)

- The installation program has been enhanced to allow the booting and installing of the OS from a USB memory stick.
- The Linux® kernel has been upgraded to the version 2.6.34 and all required drivers have been integrated to support the latest hardware.
- Several minor bugs have been fixed in the Administration Web GUI.
- The default serial speed has been changed to 115200.
- Some optimisation has been made to reduce the CDROM image size.

Version 5.6.0 (18 March 2010)

- An AntiMalware (Virus, Trojan, Worm) has been added to the appliance.

Version 5.5.5 (28 February 2010)

- The Administration Web GUI has been enhanced to allow direct accesses to menu boards from the main bar menu.

Version 5.5.4 (16 February 2010)

- The tuner module has been enhanced to manage parallel web requests more adequately.
- The guarding module has been enhanced to allow or deny the usage of direct IP addresses instead of domain names.
- The Auditing Administration Web GUI has been fixed to print messages properly.
- An anti-malware has been added to the appliance in beta test mode.
- The backup retention policy for logs has been changed so the system backs up logs for a period of 30 days.
- A new feature has been added to the system so unwanted web access and rejected requests to protected web servers are all logged in separated files.

Version 5.5.3 (15 December 2009)

- The Administration Web GUI has been fixed to properly refresh logs when an explicit refresh is invoked.
- A new option has been added to the Administration Web GUI to clear the persistent web cache.

Version 5.5.2 (30 November 2009)

- The Administration Web GUI has been fixed to properly display the top main menu in ie8.

Version 5.5.1 (16 November 2009)

- The guarding feature has been reinforced so that web users are no longer allowed to directly use IP addresses instead of domain names to bypass URL filters.
- In the Administration Web GUI, clear passwords has been removed from displayed reports.
- The Administration Web GUI has been enhanced to support IE8.

Version 5.5.0 (13 October 2009)

- The **user** command has been removed and replaced by the argument **user** added to the **admin** command.
- A new command named **cache** has been added to the system. This command allows management of some cache parameters.
- The **forceloadurl** command has been removed and replaced by the argument **loadurl** added to the new **cache** command.
- The argument "denyurl" has been added to the **filter** command. This argument allows you to set a specific URL to redirect to when an HTTP request is blocked.
- The filter and compress modules have been improved to support accessing websites that use NTLM / SSPI authentication (even if NTLM/SSPI is not compliant with HTTP).
- The URL blacklist auto updating module has been enhanced to properly download all remaining files since the last update process.
- The file transfer module has been improved to manage errors during file transfer.
- The Administration Web GUI has been modernized and improved.
- The User's Guide has been enhanced.
- USB keyboards are now supported.

- Some internal minor bugs have been fixed.

Version 5.4.2 (15 March 2009)

- An option to manage SSL CA chain has been added to the **rweb** command.

Version 5.4.1 (22 February 2009)

- The syntax of the **guard** command has been changed and new guard management features have been added to the appliance. An option allows you to update an existing blacklist category from a diff file. A second option allows you to automatically update a blacklist category since the last update/create date until today. It is also possible to program automatic blacklist category updates. Also the blacklist category save option has been removed.

Version 5.4.0 (2 January 2009)

- An LDAP authentication mode has been added to the appliance.
- The bug that was preventing the connection to internal NTP servers has been fixed.

Version 5.3.7 (25 Nov 2008)

- Now the multi CPU mode is activated during the installation if there is more than one installed CPU.
- A Huge Memory management mode (RAM > 4GB) is now available on the standard CDROM and can be chosen during the installation.

Version 5.3.6 (20 Nov 2008)

- The crash management module has been enhanced.
- The bug in the Health Checking module that inadvertently restarted services has been fixed.
- Now the **rweb** mode is turned off by default.
- An option to cancel the running **apply** operation has been added.
- The patching module has been completely reviewed.
- The Auditing Administration Web GUI has been enhanced.
- Generic content filtering rules have been updated.
- The Auditing Administration Web GUI has been fixed to properly displays all warning messages.
- The reverse web mode has been fixed to properly work even if there is only one declared HTTP website name.
- The reverse web mode has been fixed to properly work even if there is no DNS declared.

Version 5.3.5 (16 Sept 2008)

- Some internal minor bugs have been fixed.
- The CacheGuard License has been upgraded to the version 1.2. Now you can edit and modify the proprietary part of CacheGuard for your exclusive personal use. You still may not, except as permitted by applicable law, loan or create derivative works from the proprietary part of CacheGuard (see the new license).

Version 5.3.4 (28 Aug 2008)

- A CSS (Cascading Style Sheets) was added to the Administration Web GUI.
- SSL v2 is no longer supported when the appliance acts as a reverse Web proxy (only SSL v3 and TLS v1.0 are supported now).

Version 5.3.3 (29 May 2008)

- In the Administration Web GUI, the **logout** screen properly displays all images.

Version 5.3.2 (8 May 2008)

- The connection to the Auditing Web GUI works properly when the Guarding mode is deactivated (concerns only appliances installed for less than 20 users).

Version 5.3.1 (20 March 2008)

- The HTTP Transparent and HTTP Compress combination mode problem that produces some inconsistent HTTP requests has been fixed.
- Synflood rules are less aggressive so overloaded web browsing works properly without faulty rejects.
- Textual output has been formatted to comply vt100 terminals.
- The power-off button on SPC appliances works now and shuts down the system properly.
- The LCD display on SPC appliances works properly.
- The **conf diff** command has been optimised.
- A "Show Configuration" option has been added to the Administration Web GUI.

Version 5.3.0 (14 March 2008)

- The furtive error while adding a list item in the Administration Web GUI has been corrected.
- Connections to next peers work properly.
- Object sharing between cache peers has been optimised.
- All source codes are rebuilt using gcc v4.1.2.
- All basic packages have been upgraded.
- The **halt** command can power off the system even if the administrator is remotely logged in.
- The support of old Pentium Pro CPU has been added to the Linux® kernel.

Version 5.2.8 (23 December 2007)

- The memory usage has been optimised.

Version 5.2.7 (1 December 2007)

- The number of parallel connections from peers is not restricted. Peers are considered as trusted parties that do not generate flooding traffic.
- The free trial version for more than 10 users has been limited to 15 days. When the trial period is about to end, the **apply** command no longer applies a new configuration unless a valid license key is installed.

Version 5.2.6 (24 November 2007)

- In Anonymous mode, the "WWW-Authenticate" header is no longer hidden.

Version 5.2.5 (5 November 2007)

- A Synflood guarding has been added for traffic labeled **other**.
- The number of parallel connections per client IP address has been restricted, which allows this release to stop flooding.
- Bug fix: The log rotation process has been fixed to save logs with the correct date and time.
- Bug fix: The IP address configuration has been fixed when the HA mode is deactivated.
- This is the first stable version.

Version 5.2.4b (1 November 2007)

- The Synflood guarding module has been enhanced for web traffic.
- The Linux® kernel has been upgraded to 2.6.23.1.

Version 5.2.3b (26 October 2007)

- Multiple reversed HTTP websites may be associated to the same public IP address.
- The brute force attack guarding module has been enhanced for web traffic.

Version 5.2.2b (21 October 2007)

- The Auditing Administration Web GUI is activated even if the **filter** and **rweb** modes are not activated.
- In the **rweb** command, when adding a reversed website name, a mandatory IP address must be given for a HTTP website as well as for a HTTPS website.
- The QoS policy for a reversed website has been changed to be based on its public IP address.
- Several minor bugs have been fixed in the QoS module.

Version 5.2.1b (12 October 2007)

- The Administration Web GUI has been optimised.
- Bug fix: The configuration loading works properly even if the file to load does not exist.
- The reverse Auditing Web GUI documentation has been enhanced.
- Passwords having a length of 9 or greater are supported.
- FTP and TFTP protocols are supported by the Firewall.
- In High Availability mode all services are activated properly after configuration changes.

Version 5.2.0b (5 October 2007)

- X-Forwarded-Host, X-Forwarded-Server are removed from HTTP headers requests - X-Forwarded-For is also removed if no Next Peer is declared when the anonymous mode is activated.
- Port numbers for Next Peers can range from 0 to 65535 (see the **peer** command).
- An Auditing Web GUI has be added to the OS to audit the web content filtering module. Auditing allows the inspection of HTTP request content and facilitates the filtering rule design process (see **admin**, **filter** and **port** commands).
- A "Logout" link has been added to the Administration Web GUI.
- Deleting an administrator user works properly.

Version 5.1.2b (1 October 2007)

- The Via header is removed from all requests even if the anonymous mode is not activated.

- In the command **port**, the keyword "webadmin" was renamed to **wadmin**.
- In the command **password**, the keyword "webadmin" was renamed to **wadmin**.
- In the **rweb** and **transaction** commands, the keyword "print" has been renamed to **show**.
- Bug fix: The ftp passive mode can now be activated properly.
- The administration access topology can now be configured with the **admin** command.

Version 5.1.1b (22 September 2007)

- Bug fix: the website deleting with the **rweb** command works properly and all related custom filters are removed.
- Bug fix: custom filter rules are properly applied to the running configuration and appropriate services restart.

Version 5.1.0b (20 September 2007)

- TRACK and TRACE methods are denied for the embedded web server and all hosted web servers even if the filtering mode is not activated.
- Content filtering is only applicable in reverse websites and does not affect the forwarding proxy.
- Custom content filtering based on regular expressions is operational.
- The syntax of the **guard** command has been changed.
- The **conf** command is optimised to run faster.

Version 5.0.0b (9 September 2007)

- The content filtering mode (**filter** mode) for reversed websites is operational. When the **filter** and **rweb** mode are activated, requests on protected websites are filtered for generic attacks (xss, sql injection...), protocol violations and other anomalies.
- The content filtering is hardened for the Administration Web GUI.
- The configuration is properly saved for backend servers associated to a website.
- Guard categories are created even if the **guard** mode is deactivated.
- Guard black and white lists are loaded properly (the given file name must not include ".domains", nor ".expressions" nor ".urls" nor the ".gz" extensions).
- Setting VRRP in the Administration Web GUI works correctly (a wrong content filtering rule was previously set in error).

Version 4.1.6b (2 September 2007)

- By default Route Tracing (traceroute) is allowed from the internal zone to the external zone.
- Bug fix: The Administration Web GUI for the firewall configuration (Menu items "Security/External Firewall" and "Security/Internal Firewall") was fixed to work properly for long content.
- The content filtering for the Administration Web GUI is more permissive for punctuation characters.
- Some other minor bugs were corrected.

Version 4.1.5b (28 August 2007)

- The licensing is also based on the number of websites to reverse.
- The "Hard Factory Reset" procedure resets properly the "admin", "superadmin" and the root passwords.
- Images in the User's Guide available from the Administration Web GUI are shown properly.

Version 4.1.4b (22 August 2007)

- The network installation and its documentation are improved (Mainly: the TFTP IP Address is guessed and if the installation fails, the installation environment is properly reset to give the ability to relaunch the installation).

Version 4.1.3b (17 August 2007)

- Bug fix: The port forwarding integrity is properly checked during the **apply** operation (Cannot NAT the destination IP to the appliance itself).
- Bug fix: When adding firewall rules using Administration Web GUI, an empty entry does not add an "any to any" rule. To specify an "any to any" rule the keyword **any** must be specified for the Source IP, the Destination IP or the Ports field.
- Bug fix: The QoS/Incoming Flows menu item works properly in the Administration Web GUI (Bug due to contenting filtering in the Administration Web GUI).
- Bug fix: Website adding works properly in the Administration Web GUI (Bug due to contenting filtering in the Administration Web GUI).

Version 4.1.2b (13 August 2007)

- Bug fix: Network traffic other than Proxy traffic (HTTP, HTTPS and FTP) are shaped properly without abnormal slowdown.

Version 4.1.1b (10 July 2007)

- The Appliance could be installed properly using a PXE network device. The TFTP server IP address is configurable during installation.

Version 4.1.0b (9 July 2007)

- The Administration Web GUI security has been improved.
- Bug fix: Native IP addresses could be setup properly in the Administration Web GUI.
- The **rweb** VLAN is configurable using the Administration Web GUI.
- The reverse mode is configurable using the Administration Web GUI.
- The keyword "confcert" was renamed to **genssl** (related commands: **rweb** and **admin**).
- When an HTTPS reverse website is deleted, the associated host list is erased only if no other external IP address is associated with this HTTPS website.

Version 4.0.0b (24 June 2007)

- A **reverse** mode is at last available in this version. This mode allows you to implement the appliance as a reverse proxy in front of web servers to secure, accelerate and shape web traffic. (see the **mode** and **rweb** commands).
- SSH key loading works properly.
- SATA storage controller are supported again in this version (support was accidentally removed from the previous version).
- The keyword "gencert" is renamed **confcert** in the **admin** command.

Version 3.5.0b (4 June 2007)

- The QoS bandwidth shaping works properly for all types of traffic.
- The syntax of the **qos** command has changed.
- The QoS management can be deactivated using the **mode** command.
- The "fw" command has been renamed to **firewall**.
- This is an intermediate version before a main one supporting the **reverse** mode.
- The reverse mode is named **rweb** and some related commands are already integrated in the present version (but the **rweb** mode is not yet operational):
  - The reverse mode could be activated using the command: **mode rweb on**.
  - The forward mode could be deactivated using the command: **mode web off**.
  - A new vlan named **rweb** is available for web servers.
  - A Filtering mode is integrated to inspect inside web requests (see the **mode filter** command).
  - Allowed web servers can be restricted to those declared with the **access rweb**... command.

Version 3.4.0b (2 May 2007)

- The certificate generation procedure for the Administration Web GUI supports white spaces in entries.
- The alter image mode is no longer supported - The core proxy module has changed.
- Time & Date can be setup properly by using the Administration Web GUI.
- The log rotation procedure properly deletes logs older than 10 days (or with a serial number greater than 10).

Version 3.3.2b (12 April 2007)

- The documentation of the **mode** command has been fixed (gateway has been renamed to router).
- The Administration Web GUI is enhanced for the General Feature and Network related modes.
- The Administration Web GUI can show the last apply report even if the configuration is locked.

Version 3.3.1b (10 April 2007)

- In the **mode** command, **gateway** has been renamed to **router**.
- The integrated DHCP server may be activated via the CLI or Administration Web GUI.
- The integrated DHCP server supports a failover mode.
- Network PCMCIA cards are detected.

Version 3.3.0b (28 Mar 2007)

- System and access logs are rotated together even if the access log is empty.
- A VRRP IP address can be associated to the external network interface (Useful for incoming connections via the external network interface, crossing the embedded firewall and destined to internal networks).
- The access to the embedded DNS is allowed.
- In HA mode, the vrrp multicast is allowed for all IP in the local network (and not only for declared HA peers).
- In HA mode, if the health checker cannot properly restart all vital service, a fail over is forced. The forced fail over is logged in the daemon.log log file.
- When defining administrator access with the **access** command, an optional netmask could be specified.
- Bug fix: The configuration difference is correctly displayed in the Administration Web GUI.
- The Administration Web GUI is available via the embedded Proxy only when the VLAN mode is deactivated.

Version 3.2.7b (21 Mar 2007)

- Now, the Health Checker is correctly launched and checks all activated services.
- The Administration Web GUI is available via the embedded Proxy.
- Minor enhancements and optimisation.

Version 3.2.6b (14 Mar 2007)

- Bug fix: Now, the tftp command is found during the installation phase.

Version 3.2.5b (10 Mar 2007)

- When loading/saving guard categories, the category type may be optionally specified.
- Security was fixed so that, in VLAN mode, the embedded Firewall allows or denies only traffic to or from the **web** VLAN.
- The syntax of the **access** and **fw** commands has been changed. The access type **other** in the **access** command has been replaced by the **fw** command followed by the keyword **intern**. In the **fw** command, the source IP address and optionally the network mask is specified.
- Other minor bug corrections.

Version 3.2.4b (21 February 2007)

- An optional port number may be defined when adding a Next Peer.
- Support has been added for the SCSI Message Fusion Driver (required for VMware® certified version: LSI Logic).

Version 3.2.3b (13 February 2007)

- Support for TFTP to exchange Files with the appliance. To do that, the syntax of the following commands is changed: **access**, **vlan**, **conf**, **system**, **log**, **guard**.
- The completion for the **dns** command supports the keyword **localhost**.
- To respect the command syntax homogeneity, the keyword "snmp" is renamed to **mgt** for the following commands: **access**, **vlan**. The **mgt** keyword specifies "snmp" and other possible management protocols later (The snmp agent is still not integrated in this version).
- The configuration cannot be applied if the internal and external IP addresses belong to overlapped networks (The text of the error number 203 is also modified).
- The **ip** command checks if the given IP address is a valid host IP address (The network and broadcast IP address cannot be given now).
- Bug fix: Swapping between the VLAN mode and Native Mode (mode vlan on/off) restarts adequate services to bind to appropriate network interfaces.
- Bug fix: The system patching (Menu item "File/System Patches") works correctly in the Administration Web GUI now (the "Do Operation" produce the awaited result).
- Other minor bug corrections.

Version 3.2.2b (2 February 2007)

- A shortcut "Apply" button was added to the Administration Web GUI's main menu.
- The keyboard selection during installation was enhanced.
- The **access** command documentation was enhanced.
- The README.txt file in the VMware® virtual machine version package was enhanced.

Version 3.2.1b (23 Jan 2007)

- The **apply** command can be applied after a "factoryreset" without adding a DNS server.
- The Administration Web GUI is now compliant with IE7 and FireFox® 2.0.

Version 3.2.0b (17 Jan 2007)

- Initial public announcement

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Getting Started

CacheGuard-OS is an autonomous Operating System (OS) that turns an x86/x64-based machine (virtual or bare metal) into a network Security and Optimisation appliance. **Caution:** installing CacheGuard-OS on a machine will format all disks on that machine, permanently deleting all existing data.

CacheGuard-OS is built on a Linux kernel and several widely used open-source components, all compiled from source to ensure maximum integrity and security. The combination of these components and proprietary software developed by CacheGuard Technologies forms CacheGuard-OS. Note that CacheGuard does not depend on any specific Linux distribution, as it is itself an appliance-oriented Linux distribution.

The open-source software used by CacheGuard-OS is mainly distributed under the GNU GPL. Open-source programs developed by CacheGuard Technologies are distributed under the CacheGuard Licence, which is a dedicated open-source licence. Please read the Licence Agreement carefully before using the product.

The installation programme provides two installation options: Gateway or Manager. A Gateway system delivers network Security and Optimisation services. If you deploy multiple Gateways, you can manage them individually or centrally using a Manager system. In that case, install CacheGuard-OS as a Manager on a dedicated machine to centrally manage multiple Gateways. Note that only one installation type can be selected per machine - a single system cannot function as both a Gateway and a Manager at the same time.

A configured Gateway can operate in forwarding mode to protect internal users accessing the internet, and in reverse mode to secure and optimise traffic exchanged with web applications. Both modes can be enabled simultaneously on the same Gateway.

Machine Requirements

Gateway Machine Requirements

The required machine resources for a Gateway system mainly depend on the total number of users to be protected (in forwarding mode) or the maximum number of **simultaneous** users of web applications to be protected (in reverse mode). By “simultaneous users”, we mean active users who consume the appliance’s resources (mainly RAM and CPU). Simultaneous users differ from concurrent users, who may all be connected to protected web applications without necessarily consuming any resources on the CacheGuard appliance. On web application servers, concurrent users typically consume RAM and/or disk space, whereas on CacheGuard, only simultaneous users consume RAM and CPU (number of simultaneous users ≤ number of concurrent users).

During installation, the OS is fine-tuned according to the expected number of users to be supported, ensuring an optimal balance between performance and resource consumption. To provide an optimised quality of service, the tuning process assumes that not all users are connected simultaneously — only about 20 percent of them. For example, an appliance installed for 100 users is configured to support 20 simultaneous users while still allowing short bursts of up to 100 simultaneous users for limited periods.

For 100 users (20 simultaneous users), a typical machine configuration is as follows:

- Architecture: x86/x64
- CPU: 4 cores
- RAM: 8 GB
- Disk: 250 GB
- Network: 2 × Ethernet NIC

For higher numbers of users, it is advisable to use a machine with additional RAM, CPU cores, and disk capacity. As a general rule, add one CPU core and 1 GB of RAM (+75 GB of disk space in forwarding mode) for every 10 additional simultaneous users. For instance, an appliance supporting 40 simultaneous users would require six CPU cores, 10 GB of RAM, and approximately 400 GB of disk space in forwarding mode.

On physical hardware, CacheGuard-OS performs more efficiently with multiple lower-capacity disks configured as a RAID array rather than a single high-capacity disk. CacheGuard-OS natively supports software RAID using only about 3% of CPU resources. Supported RAID levels include RAID 0 (striping), RAID 1 (mirroring), RAID 5 (striping with parity), RAID 6 (striping with double parity), and RAID 10 (striping with mirroring).

CacheGuard-OS allows all integrated security and optimisation features to be enabled simultaneously on the same machine. Some features, such as real-time HTTP compression and antivirus, are more CPU-intensive than others, and activating antivirus functionality requires approximately 2 GB of RAM. The configuration guidelines provided above apply when all features are enabled concurrently. Fewer resources may be sufficient if not all features are required.

CacheGuard-OS requires at least **two NICs** (Network Interface Cards). If your machine has only one NIC, you may use

a USB Ethernet adapter as the second interface. To take advantage of link bonding or to use an auxiliary network interface, additional NICs (or USB Ethernet adapters) are required.

CacheGuard-OS can also be installed for a minimal number of users on a compact computer. The minimum configuration to support 10 users in forwarding mode is as follows:

- Architecture: x64
- CPU: 2 cores
- RAM: 4 GB
- Disk: 40 GB
- Network: 2 × Ethernet NIC

This configuration allows all CacheGuard-OS features to be enabled simultaneously on a 64-bit machine. However, if high-memory-demand services such as antivirus are not required, CacheGuard-OS can operate with as little as 512 MB of RAM. With that amount of RAM, your CacheGuard-OS-based system can function effectively as a firewall and VPN server.

Manager Machine Requirements

The required resources for a Manager system mainly depend on the total number of Gateway systems to be managed. To manage up to 10 Gateway systems, a typical configuration is as follows:

- Architecture: x86/x64
- CPU: 2 cores
- RAM: 1 GB
- Disk: 25 GB
- Network: 1 × Ethernet NIC

To manage additional Gateways, it is recommended to use a system with increased disk capacity. As a general guideline, add approximately 25 GB of disk space for every 10 additional Gateway systems to be managed.

Machine Compatibility

CacheGuard-OS supports most commonly used x86/x64-based devices available on the market. If your device is not detected during installation, please contact us - we will do our best to integrate the required drivers into the OS to ensure compatibility.

OS Installation

The same installation CD-ROM can be used to install the appliance either as a Gateway system or as a Manager system. Simply follow the on-screen instructions during installation to select the desired system type.

CD-ROM Drive Installation

To install CacheGuard-OS from a CD-ROM drive, follow these steps:

- Download the appropriate x64 or x86 ISO image file.
- Burn the ISO image to a CD-ROM using your preferred disc-burning tool.
- Boot your target appliance from the CD-ROM and follow the installation instructions.

USB Memory Stick Installation

To install CacheGuard-OS from a USB memory stick, follow these steps:

- Download the x64 or x86 ISO image file.
- Create a bootable USB stick from the ISO file. You can use UNetbootin or run the following Linux command:
  sudo dd if=CacheGuard-OS.iso of=/dev/sdX conv=fdatasync status=progress
  Replace *CacheGuard-OS.iso* with the downloaded ISO file and */dev/sdX* with the device path of your USB stick.
  CAUTION: carefully identify your USB device path to avoid erasing your workstation's disks.
- Boot the target appliance from the USB stick and follow the installation process.
- To avoid conflicts with other USB storage devices, plug in the installation USB stick last. This helps the installer distinguish it from other devices.

Network Installation

To install CacheGuard-OS over a network, the target machine must support PXE boot. An installation server is also required, running the following services:

- A DHCP server
- A TFTP server

To perform a network installation:

- Copy the cacheguard directory from the installation CD-ROM to the root directory of your TFTP server (usually /srv/tftp on Linux).

- Copy the cacheguard-boot directory from the CD-ROM to the TFTP root.
- Copy the isolinux directory from the CD-ROM into the cacheguard-boot directory on your TFTP server.
- For example, on a Linux system with the CD-ROM mounted at /mnt/cdrom and TFTP root at /srv/tftp, run:
    - sudo cp -rf /mnt/cdrom/cacheguard /srv/tftp/
    - sudo cp -rf /mnt/cdrom/cacheguard-boot /srv/tftp/
    - sudo cp -rf /mnt/cdrom/isolinux /srv/tftp/cacheguard-boot/
    - Update your DHCP configuration as follows:

```
...
allow booting;
allow bootp;
filename "/cacheguard-boot/isolinux/pxelinux.0";
subnet <network-ip-address> netmask <network-mask> {
    range <first-ip-address> <last-ip-address>;
    next-server <tftp-ip-address>;
}
...
```

    - For BIOS-based systems, use /cacheguard-boot/isolinux/pxelinux.0 as the filename. For UEFI systems, use /cacheguard-boot/pxeboot.efi.
- Finally, boot the target machine from its PXE-enabled network interface (preferably the first one).

The OVA & VHD Distribution Forms

The Gateway OVA (Open Virtual Appliance) version uses two network interfaces, which should be connected according to your configuration requirements. The VHD (Virtual Hard Drive) version can be attached to a virtual machine with at least two network interfaces. Typically, the CacheGuard **external** interface connects to an internet router or DMZ, while the **internal** interface connects to the LAN.

After booting the VM, log in as *admin* (default password: *admin*) and follow the setup procedure.

VMware® Notes

For security reasons, VMware Tools are not installed and cannot be installed on a CacheGuard appliance.

Oracle VirtualBox® Note

CacheGuard is fully compatible with Oracle VirtualBox®.

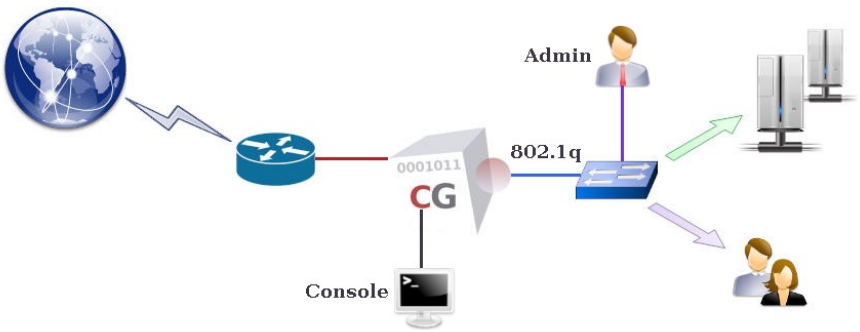
Microsoft Hyper-V® Note

When installing CacheGuard on Microsoft Hyper-V®, ensure that MAC address spoofing is disabled on the virtual machine.

Gateway Connection

In a standard configuration, CacheGuard Gateway divides your network into two zones: an external, untrusted zone (connected to the internet) and an internal, trusted LAN zone.

CacheGuard uses two logical interfaces: **external** and **internal**. Each logical interface must map to at least one physical network interface.



Use the link command (without arguments) to list detected network interfaces, or link bond to view interface associations. By default, *eth0* is assigned to the external interface and *eth1* to the internal interface.

Connect the internal interface to your LAN and the external interface to your WAN (typically your internet router).

Gateway Configuration

First Configuration

To begin, connect to the CacheGuard console interface and log in as the *admin* user.

The console can be accessed via:

- A directly attached monitor and keyboard
- An RS232 serial port (configured as 115200 8N1)

Upon first login, the setup command is automatically launched to guide you through basic network configuration. Alternatively, you can use CLI commands directly:

- *ip external 192.168.1.1 255.255.255.0*
- *ip internal 10.20.0.254 255.255.255.0*
- *ip route add default 192.168.1.254*
- *dns add 127.0.0.1*

Apply the configuration using:

- *apply*

Basic Configuration

Configuration in CacheGuard-OS is transactional: settings are defined first, then committed using the apply command. This ensures stability and allows pre-validation before activation.

The apply command verifies consistency before deployment, taking between 5-180 seconds depending on hardware and operation type. It runs in the background, allowing other commands to be executed (except configuration changes). Progress can be checked using apply report.

Accurate timekeeping is essential, as many services depend on the system clock. To set the time and date:

- *clock <YYYY/MM/DD-hh:mm:ss>*
- *apply*

You may also use NTP servers - refer to Date & Time for details.

Only trusted administrators can remotely manage the appliance. Use access to define trusted networks. Example:

- *access admin add internal 10.20.0.0 255.255.255.0*
- *apply*

To enable SSH and web administration interfaces:

- *admin ssh on*
- *admin wadmin on*
- *apply*

Connect remotely using SSH: *ssh admin@10.20.0.254*
Or via the Web GUI at: ***https**://10.20.0.254:8090*

Verify the SSL fingerprint with tls server fingerprint default. For enhanced security, set distinct passwords for console/SSH and web administration using password login.

Features (or "modes") can be enabled or disabled with the mode command. By default, both **web** (proxy) and transparent modes are enabled. Transparent mode intercepts HTTP traffic without browser configuration, while non-transparent mode requires proxy settings (e.g. *10.20.0.254:8080*).

Enable caching with:

- *mode cache on*
- *apply*

Refer to the General Modes section for a summary of available features. To secure web servers, enable reverse mode (*mode rweb on*) and configure it via rweb. To disable unused features (e.g. forwarding proxy), use mode web off.

For command assistance, use help or *help <command>* (e.g. *help ip*). Press <TAB> to autocomplete or list available parameters.

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Administration Interfaces

This section describes how to connect to a CacheGuard appliance in order to configure and administrate it. Three configuration and administration interfaces are available in a CacheGuard appliance:

- The console port
- The HTTPS interface (Web Administration GUI)
- Secure Shell (SSH)

Each interface is detailed below.

Console Port

The console port is the main interface of a CacheGuard appliance. It can either be a serial port (male DB9 RS232) or a screen and keyboard directly connected to the appliance. To use the serial port, connect the CacheGuard appliance serial port to your workstation's serial port using a crossed serial cable. You can then use your preferred terminal emulator (PuTTY, Minicom, Screen, etc.) to connect to the CacheGuard appliance. Serial communication with a CacheGuard appliance should use the following settings: 115200 8N1 (115200 baud rate, 8 data bits, no parity, 1 stop bit).

The console port is the most secure and reliable administration interface of a CacheGuard appliance. After installation, the only available administration interface is the console port. To grant remote administration access to an administrator for the first time, you must use the console port ("access admin" command).

An administrator must be authenticated before connecting to the system. The "admin" user is the main administrator with the highest administration level. Other administrator users with fewer privileges can be added to the system using the admin user command. You must log in as the "admin" user for the first connection to the CacheGuard appliance. The password to use is the one set during installation (the default password for pre-installed CacheGuard appliances is "admin"). Please refer to the Administrator Users section of this documentation for more information on administrator users.

The administration and configuration process via a character interface (console port or SSH) is carried out using the CLI (Command Line Interface). To see the list of all available commands, use the help command without arguments. To obtain help on how to use a specific command, use the help command followed by the command name. If you forget the syntax of a command, the completion feature can help you find the correct syntax. Completion is available by pressing the key.

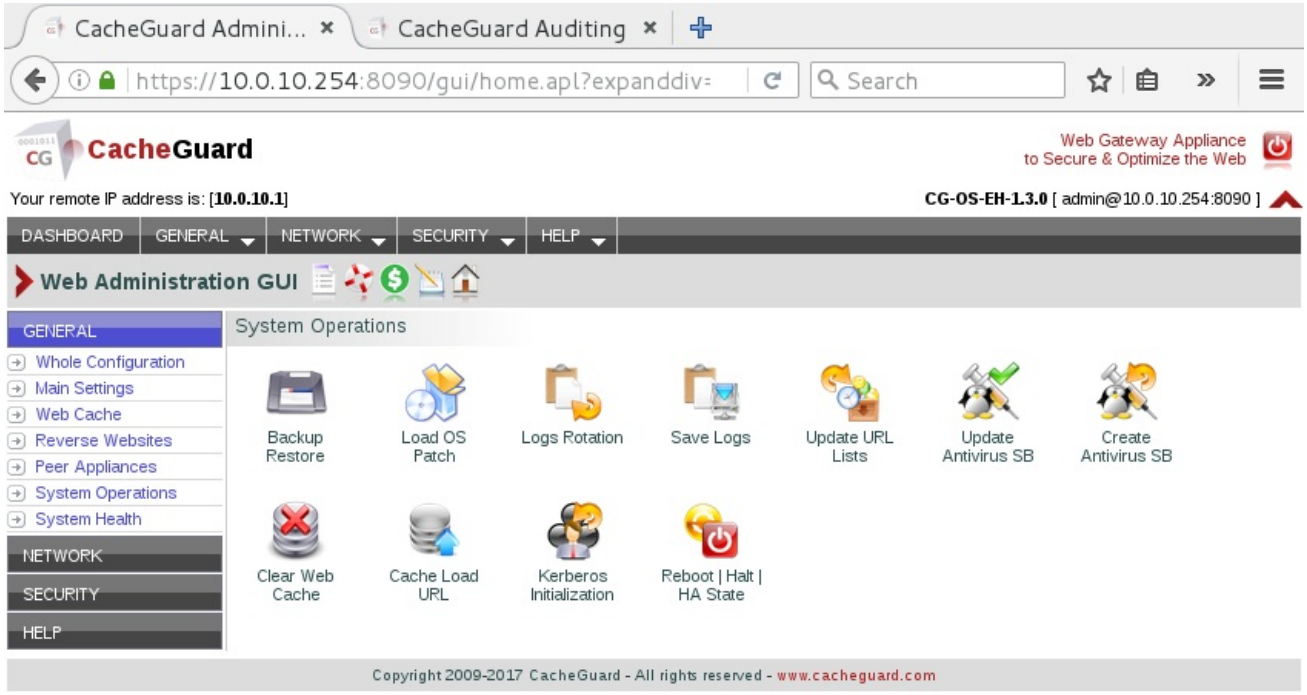
When using a character interface, the administrator has the ability to create mini-programmes using a lightweight "bash" environment (an open-source scripting language). You can use the history command to display a list of previously entered commands. To disconnect from the console port, use the exit command. The connection is also automatically closed if no command is entered for a certain period of time.

Web Administration GUI

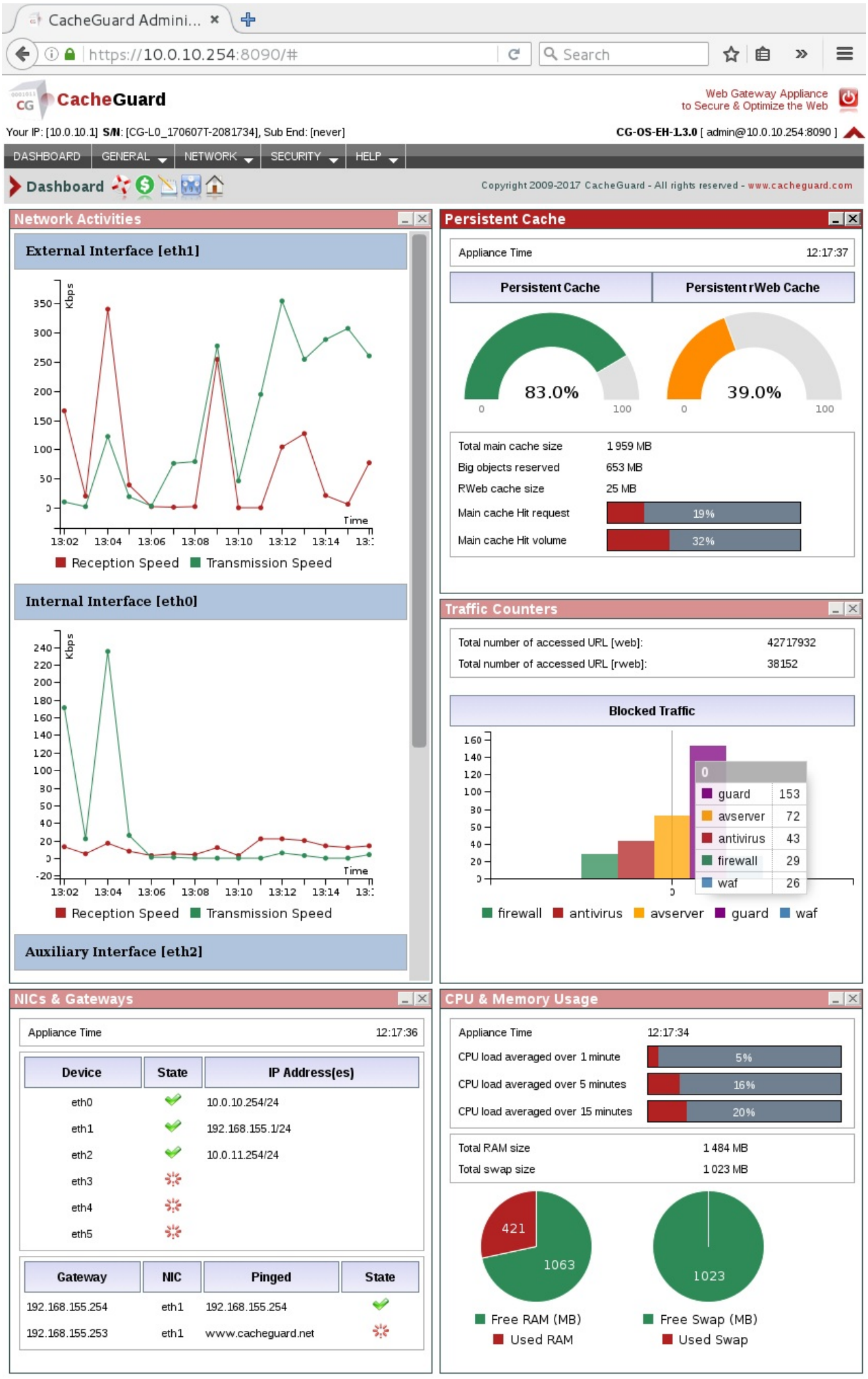
For administrators who are not familiar with a CLI (Command Line Interface) or who simply prefer a GUI (Graphical User Interface), a web administration GUI is available. To connect to the Web Administration GUI, you need a web browser. The CacheGuard appliance supports almost all modern web browsers, including (but not limited to) Firefox, Chrome, Safari, Opera and Edge. Before connecting to a CacheGuard appliance via a web browser, the Web Administration GUI must be activated on the appliance, and the remote administrator's IP address must be allowed to connect.

To activate the Web Administration GUI on the appliance and allow remote administrators in the network "10.20.0.0 255.255.255.0" to connect to the appliance via its internal network interface, use the following commands:

- *admin wadmin on*
- *access admin add internal 10.20.0.0 255.255.255.0*
- *apply*



Once the apply operation is completed, the CacheGuard appliance can be administrated using a web browser. To connect to the Web Administration GUI, use the following URL: "***https***://10.20.0.254:8090", where 10.20.0.254 is the internal IP address of the CacheGuard appliance. Note that the protocol used is HTTPS (not HTTP). The default Web Administration GUI port number is 8090. To modify this value, use the **port wadmin** command. By default, the password for the Web Administration GUI is the same as the one used to log in via the console port. It is recommended to create a separate password for the Web Administration GUI instead of using the same one as for character interfaces (using the **password wadmin** command).



The Web Administration GUI is a graphical front end to the CLI and can either replace or complement it. Using the Web Administration GUI is straightforward: first, you build a configuration using the various web pages accessible through menus, and then you activate it by performing an apply operation (as with the CLI).

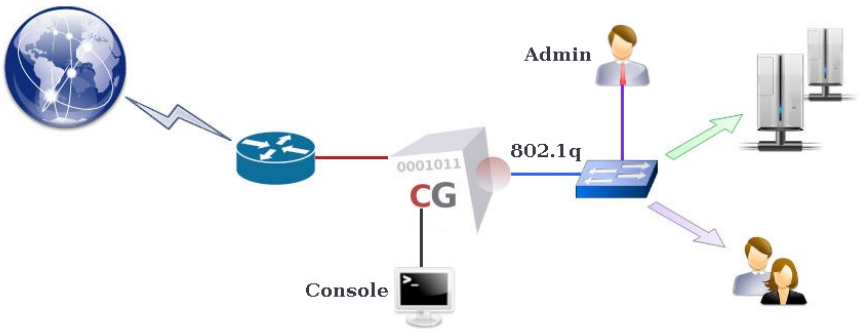
When the running (active) configuration differs from a newly built configuration, a blinking down-arrow button appears in the title bar, inviting you to click it. Clicking that icon redirects you to the Apply New Configuration page, where you can proceed with the apply operation by pressing the SUBMIT button (or the green check icon in the title bar). The Web Administration GUI is not described in detail in the User's Guide, as its usage is designed to be intuitive enough not to require extensive documentation.

Secure Shell (SSH)

The CacheGuard appliance can also be remotely administrated using an SSH client. When logged in via an SSH client, the administrator can use the CLI to administrate and configure the CacheGuard appliance. To use the SSH administration interface, you should use an SSH client installed on your workstation (for example, the Linux ssh command or the PuTTY application).

To use the SSH administration interface, remote administrators must be allowed to access the CacheGuard appliance, and the SSH service must be activated on the appliance. To activate the SSH service and allow remote administrators in the network 10.20.0.0 255.255.255.0 to connect via the internal network interface, use the following commands:

- *admin ssh on*
- *access admin add internal 10.20.0.0 255.255.255.0*
- *apply*



Once the apply operation is complete, the CacheGuard appliance can be administrated using an SSH client. To connect to a remote CacheGuard appliance from a Linux system, use the "ssh admin@10.20.0.254" command, where 10.20.0.254 is the internal IP address of the remote CacheGuard appliance. The SSH password to use is the same as the one used to log in via the console port. It is also possible to import a public SSH key into a CacheGuard appliance to allow passwordless login for administrators. Using SSH keys also enables the automation of periodic tasks, such as system backups (see the **system backup** command).

The process of generating SSH keys depends on the remote workstation used. For instance, to generate an SSH key pair (public and private) on a Linux system, you can use the "ssh-keygen" command. To import a public SSH key into a CacheGuard appliance, first copy it to a file server supporting the TFTP, FTP or SFTP protocols, and then load it into your CacheGuard appliance from that file server (which can also be hosted on your workstation). Only trusted file servers are permitted to exchange files with a CacheGuard appliance. To add the file server with the IP address 10.20.0.1 to the list of trusted file servers and allow it to exchange files via the internal network interface, use the following commands:

- *access file add internal 10.20.0.1*
- *apply*

Each SSH key must be identified with a unique identifier specified before importing it into a CacheGuard appliance. To add an SSH key identified by the "john" identifier and import it into your CacheGuard appliance from a trusted TFTP file server, use the following commands:

- *admin ssh key add john*
- *admin ssh key load john tftp 10.20.0.1 id_rsa.pub*

Here, 10.20.0.1 and id_rsa.pub represent the file server IP address and the SSH key filename on that trusted file server, respectively. After importing a public SSH key into a CacheGuard appliance, an authorised remote administrator possessing the private SSH key associated with that public key can log in without entering a password.

To disconnect an SSH session, use the exit command. The SSH session is also automatically closed if no command is entered for a certain period of time.

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CacheGuard Monitoring

CacheGuard services and the machine on which CacheGuard OS is operating can be monitored at any time to provide a quick overview of the status of a CacheGuard appliance. Monitoring enables you to detect failures and malfunctions early enough to take corrective action and maintain service continuity for your users. You can monitor the system using SNMP (Simple Network Management Protocol) or by connecting directly to the appliance and using online commands or the equivalent Web Administration GUI pages.

Note that essential services are internally monitored by a dedicated service known as the "*Health Checker*". In the event of a failure in an essential service, the "*Health Checker*" attempts to restart the failed service. This operation is logged internally, and, if configured, an SNMP manager can receive a trap notification reporting the status of the restart operation.

Direct Access

You can use several commands combined with the **report** keyword to obtain a report on the underlying function of those commands. For example, you can use the following commands:

- *system report*
- *urllist report*
- *antivirus report*

You can refer to the Commands Manual for more information on commands that support the reporting feature.

Using SNMP

A CacheGuard appliance integrates an SNMP agent that can be polled by an SNMP manager. It is also possible to configure the appliance to send SNMP traps to an SNMP receiver. Note that a CacheGuard appliance uses TCP-based SNMP traps, referred to as SNMP notifications. To allow an SNMP manager with the IP address *172.18.2.1* to poll your CacheGuard appliance, and to send SNMP notifications version 1 to a receiver with the IP address *172.18.2.2*, use the following commands:

- *admin snmp on*
- *access mon add internal 172.18.2.1*
- *admin snmp trap add v1 172.18.2.2*
- *apply*

When invoking the *admin snmp trap add v1 172.18.2.2* command, the administrator will be prompted to enter a password (the SNMP community string) to connect to the SNMP receiver. SNMP versions v1, v2c, and v3 are supported by CacheGuard OS, and the SNMP agent can be configured to operate over UDP, TCP, or TCP tunnelled over TLS. Commonly used MIBs for monitoring Linux systems are supported, as well as a dedicated MIB known as the CacheGuard-MIB. The ASN.1 MIB description for CacheGuard-MIB can be obtained via the Web Administration GUI (menu option *[NETWORK] > [Monitoring Configuration] > [SNMP Agent Settings]*), from the original installation CD-ROM, or from the official CacheGuard website.

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Traffic Logging

Logging provides visibility into allowed and denied traffic exchanged with or via a CacheGuard appliance. As an administrator, you can select which types of traffic should be logged. You may refer to the log command description in the Commands Manual to learn more about the available log types. Logs are stored locally on a CacheGuard appliance and can be saved on trusted remote file servers for inspection. You also have the option to send them in real time to remote log servers using the "syslog" protocol (based on UDP or TCP).

Managing Logs

A daily automatic log rotation during off-peak hours (between 04:00 and 06:59) backs up logs for a period of n days, where n is a value between 1 and 365 configured during the CacheGuard-OS installation. Each rotated log is identified by an integer between 1 and n, known as the log serial number. The most recent log (yesterday's log) has the number 1, and the oldest has the number n. Note that if the appliance becomes overloaded, log rotation occurs hourly in order to prevent the provisioned log space on the disks from being filled.

To save the current (today's) log, an explicit log rotation should be performed. The explicit log rotation allows you to avoid waiting for the daily automatic log rotation. To rotate logs and obtain a report on the rotation status, use the following commands:

- *log rotate*
- *log rotate report*

Once the log rotation operation is complete, you can save the required log types on a trusted file server. Note that saved logs are in gzip-compressed format. To save the most recent web access log in a file named "web-access-log.gz" on the TFTP server with the IP address 172.18.2.1, use the following command:

- *log save web 1 tftp 172.18.2.1 web-access-log.gz*

Logs can be saved only on trusted file servers. To declare a file server as trusted, use the access file command. Logging can also be completely disabled. To disable logging entirely, use the following commands:

- *mode log off*
- *apply*

Logging Web Access

All web accesses in forwarding and reverse modes can be logged. Web access logging enables you to monitor all web requests in detail and determine which machine accessed which URL at what time. To activate web access logging in forwarding and reverse modes, use the following commands:

- *mode log on*
- *log type web on*
- *log type rweb on*
- *apply*

Logging Denied Traffic

Allowed access logging concerns web traffic only (in web or rweb modes). However, all rejected traffic (web or non-web) by services running on a CacheGuard appliance can also be logged. For instance, you can configure the appliance to log all attempts to access forbidden URLs (denied by the URL guarding) and all rejected network traffic at the IP level (denied by the firewall). To achieve this, use the following commands:

- *mode log on*
- *log type guard on*
- *log type antivirus on*
- *log type waf on*
- *log type firewall on*
- *apply*

Syslog Servers

You can optionally send all locally saved logs to remote log servers that support the syslog protocol (UDP or TCP). Before activating the logging of any traffic type on a remote syslog server, that server must be explicitly allowed on the CacheGuard appliance. To configure the appliance to send its URL guarding logs in real time to a syslog server

with the IP address 172.20.2.1 using UDP, use the following commands:

- *mode log on*
- *log syslog add udp 172.20.2.1*
- *log type guard on on*
- *apply*

Multiple syslog servers can be specified on a CacheGuard appliance. In this case, logs are sent to all defined syslog servers in parallel.

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Operating System

Registration & Subscription

The registration process enables you to obtain a free serial number (S/N) that uniquely identifies your CacheGuard appliance. For paid subscription-based services, such as a support contract, the S/N is used to purchase subscriptions. After the initial subscription period, renewal is required to continue benefiting from the purchased subscriptions.

The registration process is generally manual and should be initiated from an already installed appliance. Note that a CacheGuard appliance deployed on a public cloud is automatically registered during deployment. To begin the registration process, it is recommended to use the Web Administration GUI, as you will be prompted to connect to the CacheGuard appliance registration web portal to obtain a One-Time Password (OTP).

To start the registration process, navigate to *[GENERAL] > [Main Settings] > [Registration & Subscription]* in the Web Administration GUI and follow the displayed instructions. Refer to the register command in the Commands Manual for further information.

Backup & Restore

To enable rapid recovery from hardware or software failures, the configuration of a CacheGuard appliance and its essential data (antivirus signatures, URL lists, SSL certificates, etc.) can be saved on a file server and later restored on a freshly installed appliance. This is achieved by reinstalling CacheGuard-OS on a new machine.

To back up a CacheGuard appliance, the system must first create a local backup file, which can then be saved on a trusted file server. The backup creation process runs in the background; you must wait for its completion before saving the generated file.

To create and save a system backup named "*cacheguard.backup*" on the trusted TFTP server with the IP address *172.18.2.1*, use the following commands:

- *system backup create*
- *system backup create report*
- *# Wait for the backup process to complete...*
- *system backup save tftp 172.18.2.1 cacheguard.backup*

Backup files can only be saved on trusted file servers. To declare a file server as trusted, use the access file command.

To restore a previously saved backup named "*cacheguard.backup*" from the trusted TFTP server with the IP address *172.18.2.1*, use:

- *system backup load tftp 172.18.2.1 cacheguard.backup*
- *apply*

Note that this backup and restore method is valid only if both the freshly installed and the failed CacheGuard appliances run the same CacheGuard-OS version. If they differ, a logical restore is required. The logical restore procedure is described in the Reinstalling the OS section below and requires a copy of your appliance's logical configuration.

Patching the OS

CacheGuard Technologies periodically releases new CacheGuard-OS versions and provides OS patches to upgrade existing installations. It is strongly recommended to keep your CacheGuard appliance up to date by installing the latest available patches.

Applying OS patches carries some inherent risk. Therefore, it is strongly advised to save your logical configuration and related data/files (such as SSL certificates, custom WAF rules, and antivirus whitelists) on a trusted file server before patching. This ensures that you can reinstall your appliance from scratch using the latest version and then restore your configuration. The Reinstalling the OS section below describes this recovery process.

OS patches can be loaded from a trusted file server and applied to your CacheGuard appliance (refer to the access file command to declare a server as trusted). To load a patch file named *CacheGuard-UF-64-2.1.3-patch.cgp* from a trusted TFTP server with the IP address *172.18.2.1*, use the following commands:

- *system patch tftp 172.18.2.1 CacheGuard-UF-64-2.1.3-patch.cgp*
- *apply*

CacheGuard-OS patches can also be downloaded from official CacheGuard servers on the internet and applied directly from the appliance. Refer to the system command in the Commands Manual for details.

Caution: During the patching process, ensure that the appliance has a stable power supply. If a power failure or interruption occurs during patching, or if the process fails for any reason, the appliance may become inconsistent. In such cases, the only recovery option will be to reinstall CacheGuard-OS and manually restore your configuration.

Reinstalling the OS

Some major CacheGuard-OS releases are distributed without a corresponding patch. In such cases, you must reinstall CacheGuard-OS from scratch. To simplify recovery, you can save the appliance's logical configuration and related data/files (e.g. SSL certificates, custom WAF rules, antivirus whitelist) on a trusted file server for later restoration.

To save the logical configuration in a file named "*CG.conf*" and related data in a folder named "*CGFiles*" on a TFTP server with the IP address *172.18.2.1*, use:

- *conf save tftp 172.18.2.1 CG.conf*
- *file save tftp 172.18.2.1 CGFiles*

To restore the configuration and related files, use:

- *conf load tftp 172.18.2.1 CG.conf*
- *file load tftp 172.18.2.1 CGFiles*
- *apply*

Note that there are limitations to this logical backup and restore method. The following data/files are not saved and therefore cannot be restored:

- Administrator passwords
- Antivirus signatures
- URL lists

However, antivirus signatures and URL lists are automatically downloaded from trusted servers during the *apply* operation. If you have secondary administrator users, you will need to recreate them manually.

Rebooting the Appliance

In certain circumstances, you may need to reboot your CacheGuard appliance. To reboot the appliance, use the reboot command.

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Administrator Users

Only authenticated users are allowed to administer and configure a CacheGuard appliance. The primary administrator user is named "admin". This user exists by default in the system and has the highest level of privilege.

Secondary administrator users with reduced privileges can be added to the system. Secondary administrators are allowed to read configurations but cannot modify them (only the "admin" user has this privilege). Certain secondary administrative operations, such as cache loadurl, can be executed by secondary administrators. The admin user command allows you to manage secondary administrator users.

To create a new administrator user called "john", log in as the "admin" user first and then use the following commands:

- *admin user add john*
- *apply*

When running the "admin user add john" command, the "admin" user is prompted to choose a temporary password for "john". Once the apply operation is completed, the administrator user named "john" can log in. The first time a secondary administrator logs in to a CacheGuard appliance, they are required to change their password by entering a compliant one (refer to the password command to learn more about allowed password formats).

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General Modes

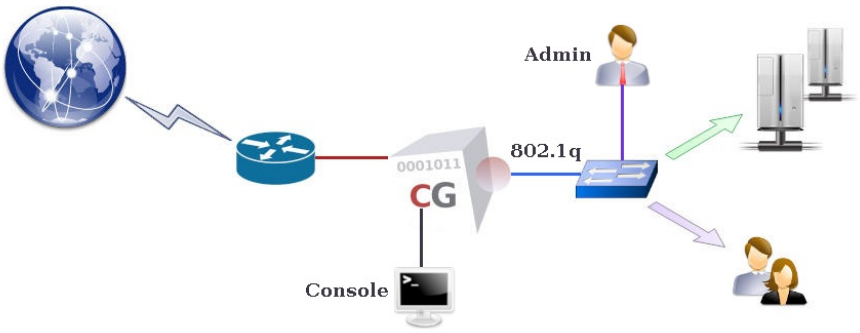
A CacheGuard appliance is an integrated system that secures and optimises internet traffic by providing a comprehensive range of features. To achieve an optimised configuration, it is recommended that you activate only the features you actually need and deactivate all others. However, if required, it is possible to activate all available features simultaneously on the same appliance, and all activated features will operate seamlessly and efficiently together.

The mode command allows you to activate or deactivate different features known as modes. For almost all modes, there is a command with the same name that allows you to configure that mode. For example, the embedded firewall can be activated using the mode firewall on command, and the firewall can be configured using the firewall command. There are two main groups of modes: network modes and function modes. The sections below provide a brief description of all available modes. Note that the apply command must be invoked after activating or deactivating a mode.

Network Modes

This section provides a brief description of all network modes. Available network modes operate at the TCP/UDP IP layer and are as follows:

- DHCP Server
- Caching DNS
- High Availability
- Passive FTP
- Quality of Service
- Network Router
- Source NAT
- Transparent Web
- Transparent Web SNAT
- Tagged VLANs



DHCP Server

A CacheGuard appliance integrates a DHCP server to dynamically assign IP addresses to connected devices. To activate the DHCP server, use the mode dhcp on command. You can configure the DHCP server using the dhcp command. This command allows you to define dynamic IP ranges and, if necessary, reserve IP addresses for specific devices.

Caching DNS

A CacheGuard appliance integrates a caching-only Domain Name Server that can be used by all other integrated services as well as by external clients. To activate the integrated DNS server and make it available for external use, execute the following commands:

- *dns raz*
- *dns add localhost*
- *mode dns on*

High Availability

Two or more CacheGuard appliances can be deployed in HA (High Availability) mode using the VRRP protocol. In addition, network interfaces (**external**, **internal**, etc.) of a CacheGuard appliance can be associated with multiple physical interfaces to provide link redundancy. To activate HA mode, use the mode ha on command. Once HA mode is

activated, use the vrrp command to associate one or more VRRP IP addresses with at least one network interface. To associate more than one physical network interface with a logical interface, use the link command.

Passive FTP

By default, a CacheGuard appliance uses the passive FTP protocol to initiate FTP sessions with external FTP servers. To use active FTP mode instead, you must deactivate passive FTP mode by using the mode ftppassive off command. Note that switching between passive and active FTP modes is done globally (it cannot be set per FTP session).

Quality of Service

A CacheGuard appliance can shape and schedule network traffic to offer QoS (Quality of Service) to users and applications. Using the QoS manager in a CacheGuard appliance allows you to reserve network bandwidth for critical applications. To activate QoS mode, use the mode qos on command. You can then configure traffic shaping using the qos command. All traffic destined for services (proxy, antivirus, etc.) running on the appliance itself, as well as routed traffic passing through it, is managed by the QoS system.

Network Router

A CacheGuard appliance can act as a network router supporting static routes and multi-gateway configurations. Use the mode router on command to activate router mode. To configure routing, use the ip route command.

Source NAT

A CacheGuard appliance can perform Source Network Address Translation (SNAT) on all outgoing traffic through its external interface, replacing the source IP with its own external IP address. This mode is called **snat** and can be activated using the mode snat on command.

Transparent Web

A CacheGuard appliance can be transparently deployed in a network to intercept web traffic for management (caching, filtering, etc.). In transparent mode, there is no need to configure web browsers to use the CacheGuard appliance as a web proxy. The simplest way to implement a CacheGuard appliance transparently is to use it as the default gateway to the internet. This mode is called **tweb** (or **transparent**) and can be activated using the mode tweb on (or mode transparent on) command. The transparent (or tweb) command can then be used to selectively intercept web traffic.

As HTTP is increasingly being replaced by HTTPS, the capabilities of transparent mode are significantly reduced. Fortunately, a CacheGuard appliance can also intercept HTTPS traffic. The transparent interception of HTTPS traffic is referred to as **sslmediate** mode and is described in the Function Modes section below. This mode is considered a function mode rather than a network mode because it involves managing SSL CA certificates in addition to network configuration.

Transparent Web SNAT

In transparent mode (**tweb**), intercepted web traffic can either preserve real source IP addresses or be Source NATed with the CacheGuard appliance's external IP address. The **tnat** (transparent NAT) mode activates (**on**) source IP address translation for transparently intercepted web traffic. The **tnat** mode is activated by default. To deactivate it, use the mode tnat off command. Note that when **tnat** is deactivated, routed web traffic must not be asymmetric with respect to the CacheGuard appliance (i.e. all incoming and outgoing web traffic exchanged with clients should pass through the same network interface).

Tagged VLANs

A CacheGuard appliance supports IEEE 802.1Q VLAN (Virtual LAN) tagging on its internal network interface to secure and isolate predefined functional traffic (**admin**, **web**, **rweb**, etc.). To activate VLAN mode, use the mode vlan on command. Once VLAN mode is activated, use the vlan and ip commands to define VLANs and assign IP addresses.

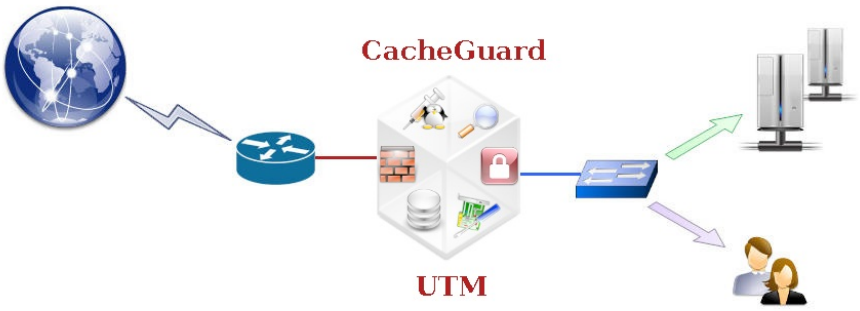
If both **web** and **rweb** modes are activated on the same appliance, and SSL offloading is configured (where HTTPS traffic between the CacheGuard appliance and real web servers is unencrypted), it is recommended to use a distinct VLAN for **rweb** traffic. Refer to the rweb command to learn how to configure the **rweb** mode for SSL offloading.

Function Modes

This section provides a brief description of all function modes. Available function modes operate at the application layer and are as follows:

- Anonymous Browsing
- Antivirus
- Web Authentication
- Stateful Firewall
- Forwarding Web Proxy
- HTTP Compression

- OCSP Responder
- Reverse Web Proxy
- SSL Mediation
- Traffic Logging
- URL Guarding
- IPsec VPN
- Web Application Firewall
- Web Caching



Anonymous Browsing

A CacheGuard appliance can modify certain HTTP headers to make web requests anonymous. To activate anonymous browsing mode, use the mode anonymous on command. Note that activating this mode may lead to being blocked by certain websites; therefore, it is recommended to keep this mode deactivated unless required.

Antivirus

A CacheGuard appliance embeds an antivirus that blocks malware (viruses, trojans, and worms) in web traffic destined for web clients (human or machine) as well as web servers. The antivirus can also be used as a service by external systems such as an MTA (Mail Transfer Agent). To activate the antivirus, use the mode antivirus on command. The antivirus command is used to configure it.

Web Authentication

The embedded web proxy in a CacheGuard appliance can request LDAP or Kerberos servers to authenticate web clients before granting them access to the web (or web servers in reverse mode). To activate authentication mode, use the mode authenticate on command. You must then configure authentication using the authenticate command.

Stateful Firewall

A CacheGuard appliance integrates a stateful firewall that enables control of routed network traffic based on source and/or destination IP addresses and protocols. In addition, the firewall supports source and destination NAT. To activate firewall mode, use the mode firewall on command. The firewall is configured using the firewall command.

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Configuration Management

A CacheGuard appliance is configured and administered using online commands or the Web Administration GUI. There are mainly two types of commands: commands with an immediate effect and configuration commands. A CacheGuard configuration is represented by a set of configuration commands invoked with the appropriate arguments.

Configuration commands can be invoked with or without arguments. Without arguments, a configuration command normally displays the configuration information related to that command. When invoked with one or more arguments, the command usually modifies the configuration. For example, the ip command, when invoked without arguments, displays all IP addresses associated with network interfaces. When invoked with a network interface name, an IP address, and a subnet mask, it modifies the IP address associated with that interface. For instance, the "*ip external 192.168.22.254 255.255.255.0*" command sets the IP address associated with the "*external*" network interface to *192.168.22.254 255.255.255.0*.

Invoked without arguments, the conf command displays the list of all configuration commands and their associated arguments that form the configuration. In CacheGuard-OS, there is always a current (or running) configuration and a new configuration. The new configuration may be identical to or different from the current configuration. If a configuration command is invoked with arguments that modify the current configuration, the new and current configurations will differ. It is important to note that invoking configuration commands does not immediately affect the running configuration.

To activate a new configuration, or in other words to replace the running configuration with a new one, the apply command must be invoked. You can use the **conf diff** command to compare the new configuration with the running configuration and display their differences.

Command Arguments

There are four types of command arguments:

- Scalar arguments (for example, an IP address)
- Keyword arguments (for example, **external**)
- Boolean parameters (valid values are **on** and **off**)
- List arguments (for example, a list of static IP routes)

A Boolean parameter is typically used to activate (**on**) or deactivate (**off**) a sub-configuration state. For example, the mode cache on command activates web caching. A list argument is managed using the following keyword arguments:

- **add**: adds an element to the end of a list.
- **insert**: inserts an element before a specified element in a list.
- **del**: deletes an element from a list.
- **raz**: clears (empties) the list.

For example, the ip route add *10.0.10.0 255.255.255.0 192.168.60.254* command adds a route to *10.0.10.0 255.255.255.0* via the *192.168.60.254* gateway to the list of static routes. The ip route del *10.0.10.0 255.255.255.0 192.168.60.254* command removes that route, and the ip route raz command removes all routes from the list of static routes.

Applying a Configuration

In CacheGuard-OS, there is always a current (or running) configuration and a new configuration. To activate the new configuration, the apply command must be invoked. The apply operation is a background process that may take from a few seconds to several minutes, depending on the nature of the new configuration and the performance of the machine running CacheGuard-OS. The apply report command allows you to obtain an execution report of the most recent **apply** operation. If, after making a new configuration, you decide to discard it, use the cancel command. The cancel command cannot be invoked while an **apply** operation is running. To cancel an ongoing **apply** operation, use the apply cancel command.

The new configuration to be applied is a set of commands that must form a consistent configuration. In other words, commands in a configuration must be compatible with one another. For example, you cannot activate (or apply) a configuration that specifies a static route via a gateway that is not within the broadcast domain of a connected network interface. The apply command is responsible for ensuring the integrity and consistency of configurations. If the new configuration is inconsistent, the apply command rejects it, the apply operation does not start, and a list of integrity errors is displayed.

Load and Save a Configuration

The current or new configuration can be saved on a file server (FTP, TFTP, etc.). Only trusted file servers are permitted to exchange files with a CacheGuard appliance. Therefore, it is necessary to trust a file server before saving a configuration on it. You must use the access command to add a file server to the list of trusted file servers. For example, to trust a TFTP file server with the IP address *172.18.2.1* for file exchange via the **internal** network interface, use the following commands:

- *access file add internal 172.18.2.1*
- *apply*

Once the apply operation is complete, you can save the current configuration to a file named *cacheguard.conf* on that server using the conf save tftp *172.18.2.1 cacheguard.conf* command. The saved file will contain a list of configuration commands. The configuration file can be reloaded into the system at any time using the conf load tftp *172.18.2.1 cacheguard.conf* command. Note that the apply command is not included in the saved file and must be manually executed afterwards.

The configuration discussed in this section is also known as the logical configuration, as the real operational configuration may include additional data or files such as SSL certificates, custom WAF rules, and SSH public keys. To save all related data to a configuration in separate files on a file server, use the file command. For example, to save all data related to a configuration in a folder named "*CGFiles*" on a TFTP file server with the IP address *172.18.2.1*, use the file save tftp *172.18.2.1 CGFiles* command. To reload those files back into the system, use the file load tftp *172.18.2.1 CGFiles* command.

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Date & Time

Many services in CacheGuard-OS, such as (but not limited to) web caching, antivirus, and URL guarding, depend on the CacheGuard appliance's internal clock. In addition, certain periodic tasks should run during off-peak hours to avoid affecting end users during business hours. That is why setting the correct date and time is essential for operating CacheGuard-OS efficiently.

It is also important to note that the current real time on a CacheGuard appliance depends on the time zone in which it operates. For example, to set the time zone to Berlin (Germany), use the following commands:

- *timezone Europe/Berlin*
- *apply*

You can use the timezonelist command to display the list of all available time zones. The internal clock can be manually configured using the clock command or automatically synchronised using remote NTP (Network Time Protocol) servers.

Manual Setting

To manually set the date and time, use the clock *<YYYY/MM/DD-hh:mm:ss>* command, where *<YYYY/MM/DD-hh:mm:ss>* is defined as follows:

- YYYY - a 4-digit year
- MM - a 2-digit month
- DD - a 2-digit day
- hh - a 2-digit hour
- mm - a 2-digit minute
- ss - a 2-digit second

For example, a valid date and time specification is: *2024/03/20-03:06:26*.

NTP Servers

To maintain an accurate clock on a CacheGuard appliance, it is highly recommended to use secure and reliable NTP servers. For security and redundancy, it is best to rely on at least three NTP servers and always use an odd number of servers from different providers. The ntp command allows you to manage NTP servers. For example, to add an NTP server with the IP address *192.168.0.1*, use the following commands:

- *ntp add 192.168.0.1*
- *apply*

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The Network

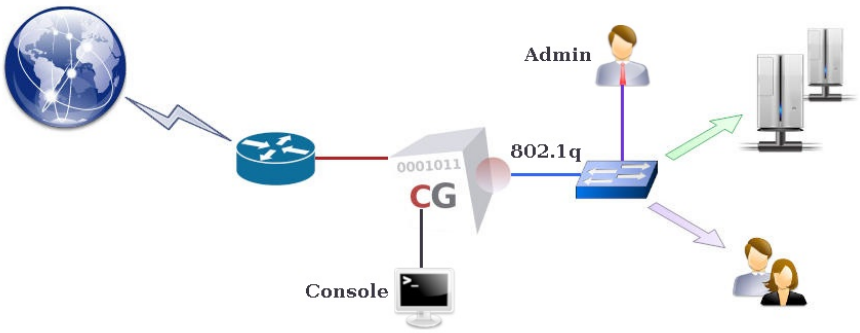
A freshly installed CacheGuard-OS has no IP configurations at all and before being able to connect to it as an administrator (**admin** user), you must set an IP address for at least one of its network interfaces. The only way to set an IP address for the first time on a CacheGuard appliance is to use the CacheGuard appliance console port. At the first login (as the **admin** user) via the console port, a command named **setup** is automatically executed and allows you to set CacheGuard's **internal** and **external** IP addresses. As an alternative method, you can use the ip command.

Network Interfaces

With a CacheGuard appliance, the network is divided into at least two zones: the *internal zone* and the *external zone*. The *external zone* is considered an untrusted zone (the internet), while protected users and servers should be placed in the *internal zone* (considered trusted). A third optional zone called *auxiliary* can be created and used as per your convenience (for instance, as a DMZ or as a Back Office zone). The *internal zone* can optionally be divided into sub-zones using tagged VLANs.

CacheGuard appliance connects each zone via a distinct logical network interface. Hence, a CacheGuard appliance can support up to three logical network interfaces: the **external**, the **internal**, and the **auxiliary** logical network interfaces. In this network topology, internal users and servers are routed via the **internal** interface, while the CacheGuard appliance uses its **external** interface to connect to the internet.

A logical network interface should be associated with at least one physical network interface. The link command allows you to associate a logical network interface with a physical (*eth0*, *eth1*...) network interface. A link configuration in which more than one physical network interface is associated with a logical network interface is called *link bonding*. The *link bonding* is an active/backup link configuration that allows service continuity in the event of a link failure. You can refer to the Link Bonding section for further information on *link bonding*.



Appliance IP Addresses

To support the IP protocol, a logical network interface must be associated with a main IP address. The ip command allows you to set main IP addresses. Additional IP addresses can be implicitly associated with a logical network interface by other commands. For instance, the vrrp command allows you to create floating IP addresses in HA mode, while the rweb command may create IP aliases associated with a main IP address. The example below sets the **external** IP address to *192.168.1.1 255.255.255.0* and the **internal** IP address to *10.20.0.254 255.255.0.0*:

- *ip external 192.168.1.1 255.255.255.0*
- *ip internal 10.20.0.254 255.255.0.0*
- *apply*

Using 802.1q VLANs

CacheGuard appliance supports 802.1q VLAN (Virtual LAN) tagging on its internal network interface to secure and isolate predefined functional traffic (**admin**, **web**, **rweb**...). When using tagged VLANs, a pseudo network interface is implicitly created for each defined VLAN. To use VLANs, you have to activate the VLAN mode by using the mode vlan on command and then configure VLANs with the help of the vlan and ip commands. Note that in VLAN mode, the native IP address associated with the **internal** network interface is no longer active, and you should configure a separate IP address for each pseudo network interface. For instance, the following commands define tagged VLANs 10, 20, and 30 respectively associated with the **web**, **rweb**, and **admin** functional traffic, and set a distinct IP address for each created VLAN (all other functional traffic will remain in the default VLAN 0).

- *vlan web 10*
- *vlan rweb 20*

- *vlan admin 30*
- *ip internal.0 10.0.0.254 255.255.255.0*
- *ip internal.10 10.0.10.254 255.255.255.0*
- *ip internal.20 10.0.20.254 255.255.255.0*
- *ip internal.30 10.0.30.254 255.255.255.0*
- *mode vlan on*
- *apply*

Network Routes

To route (or forward) IP traffic, you must create routing tables. CacheGuard-OS supports static routes only. However, you have the possibility to create multi-gateway routes to balance the IP routing between multiple gateways. In a multi-gateway configuration, gateway failures can be detected and the routing configuration dynamically modified to avoid routing traffic via failed gateways. Note that gateways in a route specification should be directly connected to a CacheGuard network interface and have an IP address in the same network as the connected network interface IP address. For example, the following commands allow you to create two default gateways via the *192.168.1.254* and *192.168.1.253* gateways and a static route to the *172.22.22.0 255.255.255.0* network via the *10.20.0.1* gateway.

- *ip route add default 192.168.1.254*
- *ip route add default 192.168.1.253*
- *ip route add 172.22.22.0 255.255.255.0 10.20.0.1*
- *apply*

Domain Name Servers

To connect to external name-based services (for instance, websites), CacheGuard appliance needs a DNS (Domain Name Server) to translate domain names to IP addresses. CacheGuard appliance embeds a caching-only DNS (Domain Name Server) that you can activate by adding the *localhost* (or the *127.0.0.1* loopback IP address) to the list of DNS servers. You also have the possibility to add external DNS to the system using the dns command. If you activate the internal DNS server, you will have the possibility to allow external clients to use it as a service. Note that you can restrict DNS access to trusted networks only by using the access command. The example below activates the internal DNS and allows IP clients to use it as a service.

- *dns add 127.0.0.1*
- *mode dns on*
- *apply*

DHCP Server

CacheGuard appliance integrates an easy-to-handle DHCP server that you can activate to deliver dynamic IP addresses to connected devices. The dhcp command allows you to define dynamic IP address ranges and/or fixed IP addresses for particular devices identified by their MAC addresses on an Ethernet network. Note that dynamic IP addresses can only be assigned to devices connected to the **internal** network interface (or the **web** interface in VLAN mode) and the **auxiliary** interface. The example below activates the DHCP server, configures it for the internal interface to deliver dynamic IP addresses between *10.20.0.11* and *10.20.0.15*, and fixes the IP address *10.20.0.10* and hostname *john* for a device having the MAC address 00:01:00:02:00:03.

- *mode dhcp on*
- *dhcp range add internal 10.20.0.11 10.20.0.15*
- *dhcp fixed add john internal 00:01:00:02:00:03 10.20.0.10*
- *apply*

High Availability

CacheGuard appliance uses several technologies to ensure the *High Availability* of the services that it offers. The HA in a CacheGuard appliance is based on redundancy and resiliency concepts. If you plan to implement CacheGuard solutions to secure and/or optimise critical services in your organisation, it is highly recommended to implement the HA offered in CacheGuard-OS.

The VRRP Protocol

By associating several CacheGuard appliances in HA mode, you can ensure service continuity and HA (High Availability) in the event of a software or hardware failure on one of them. To associate two (or more) CacheGuard appliances in HA mode, you must activate the HA mode on them using the mode ha on command. In HA mode, operational appliances automatically start handling network traffic that was initially destined for a failed appliance. The HA mode is based on the VRRP (Virtual Router Redundancy Protocol) and can be configured using the vrrp command.

When using VRRP, two (or more) CacheGuard appliances share the same virtual IP (VRRP IP) address (in addition to their real IP addresses) on their same network interface. External services or clients should then address appliances configured in HA mode by using their VRRP IP addresses (and not their real IP addresses). A VRRP IP address can be **master** (active) or **backup** on a network interface. The master VRRP IP is active on a network interface until a failure occurs on that interface. Following that failure, the backup VRRP IP becomes active (on the backup appliance). Each

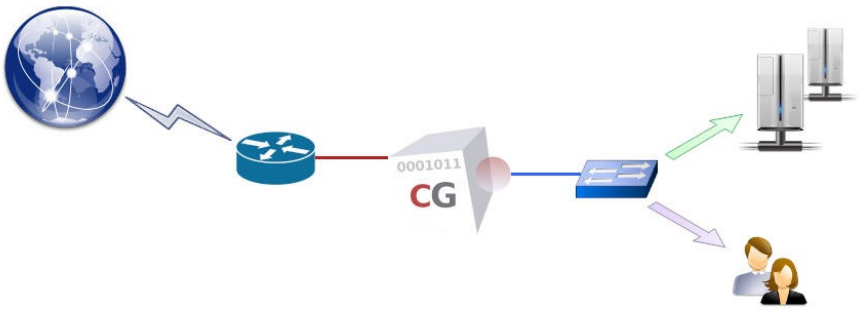
CacheGuard appliance embeds a service called health checker that continuously verifies the health of the services running on it. In case of a repeated failure on a service, the health checker deactivates its network interfaces, allowing other associated appliances in HA mode to take over the failed appliance.

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Transparent Mode

CacheGuard appliance integrates a web proxy that can be explicitly used by web clients (users or machines). In an explicit implementation, web users must configure their browsers to use the CacheGuard appliance as an HTTP/HTTPS proxy by specifying its **internal** IP address (**web** address in VLAN mode) and proxy port (8080 by default). If modifying browser configurations is not an option in your network, you can implement CacheGuard in transparent mode. In transparent mode, HTTP traffic (and optionally HTTPS traffic) is transparently intercepted by the appliance and can then be processed by the various integrated services that the CacheGuard appliance offers (such as URL filtering, antivirus protection, web caching, and more).

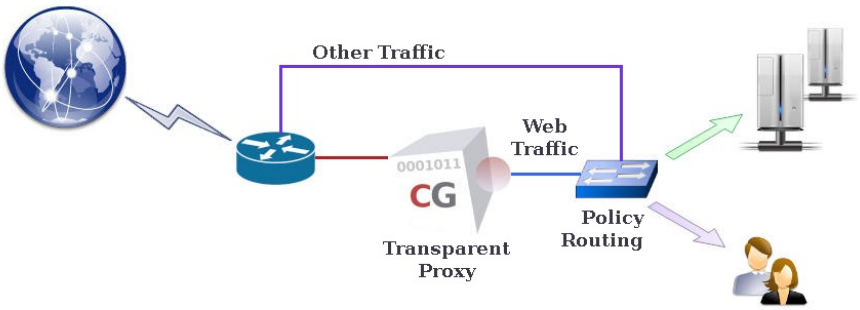
To function as a transparent web gateway (web proxy), the CacheGuard appliance must be positioned along the web traffic path - meaning that all web traffic must pass through it. The simplest method to achieve this is to use the CacheGuard appliance as the default gateway for your LAN. If using CacheGuard as the default gateway for all internet traffic is not desirable, a Layer 4 (L4) switch may be employed to implement policy-based routing, ensuring that only web traffic is routed through the CacheGuard appliance. To activate the transparent mode, use the mode tweb on (or mode transparent on) command.



Note that transparent interception of HTTPS traffic requires the SSL mediation feature to be enabled on your CacheGuard appliance. Implementing SSL mediation also requires the deployment of the CacheGuard appliance CA certificate on all web client devices. Please refer to the SSL Mediation section for guidance on implementing this functionality.

Using a Layer 4 Switch

A Layer 4 (L4) switch allows routing of IP traffic based not only on IP addresses but also on TCP/UDP headers. To ensure that only web traffic is routed through the CacheGuard appliance while other traffic is directed through a different gateway, your policy-based routes on the L4 switch should route all traffic destined for TCP port 80 (HTTP) and, optionally, TCP port 443 (HTTPS) via the CacheGuard appliance, while all other traffic continues via your standard internet gateway.



If you are familiar with Linux, you can also use a Linux machine to implement this type of policy-based routing. Numerous examples and tutorials are available online describing this configuration. As an introduction, the following example demonstrates how to implement policy-based routing on a Linux machine:

- iptables -t mangle -A PREROUTING -p tcp --dport 80 -j MARK --set-mark 5
- echo "100 transparent-proxy" >> /etc/iproute2/rt_tables
- ip route add default via <cacheguard-internal-ip> table transparent-proxy
- ip rule add fwmark 5 table transparent-proxy

In the example above, <cacheguard-internal-ip> should be replaced with the internal IP address of your CacheGuard appliance. Please refer to the ip and iptables manual pages on a Linux system for further information on these commands.

Selective Transparency

Once transparent mode is activated on a CacheGuard appliance, all web traffic routed through it is intercepted, regardless of source or destination IP addresses. This behaviour may be restrictive for users who require more control over their web traffic (for example, administrators). To address this, web traffic interception can be limited to specific networks using the transparent (or tweb) command. For instance, to limit transparent mode to web clients within the *172.18.2.0 255.255.255.0* or *10.26.0.0 255.255.0.0* networks that are routed via the **internal** network interface, use the following commands:

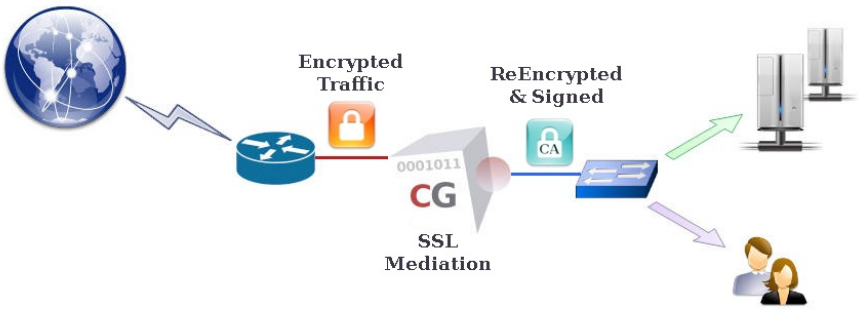
- *mode transparent on*
- *transparent raz*
- *transparent add internal 172.18.2.0 255.255.255.0*
- *transparent add internal 10.26.0.0 255.255.0.0*
- *apply*

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SSL Mediation

CacheGuard appliance can decrypt HTTPS (encrypted) traffic to enable inspection of its content. After inspection, the encrypted web traffic may be cached or blocked if malware is detected. In this way, bandwidth-intensive content such as video segments can be cached, and malicious content blocked, even when encrypted. Without SSL mediation, such traffic handling would not be possible, as caching or inspecting encrypted data cannot be performed. The purified and/or cached traffic is then re-encrypted by CacheGuard-OS before being sent to web clients. This feature is referred to as SSL mediation on a CacheGuard appliance.

Note that decrypting and re-encrypting HTTPS traffic can be regarded as a MITM (Man-in-the-Middle) attack. You should therefore ensure that enabling SSL mediation complies with the laws and regulations applicable in your country and/or organisation. It is also important to understand that enabling SSL mediation is entirely at your own responsibility. You are invited to read the CacheGuard-OS Licence Agreement for further information.



When SSL mediation is enabled, decrypted traffic is re-encrypted before being sent to web clients. To transmit re-encrypted traffic, the web proxy uses dynamically generated SSL certificates signed by a private CA (Certificate Authority) certificate. As web clients must trust the private CA certificate (referred to as the system CA certificate), it must be imported into client browsers prior to enabling SSL mediation. Otherwise, HTTPS communication will fail. You can configure whether HTTPS traffic is intercepted transparently using the sslmediate command. To enable SSL mediation and configure it to transparently intercept HTTPS traffic, use the following commands:

- *mode transparent on*
- *mode sslmediate on*
- *sslmediate transparent on*
- *apply*

The System CA

The CacheGuard CA certificate is referred to as the system CA certificate and is accessible at *http://<cacheguard-internal-ip>/*, where *<cacheguard-internal-ip>* represents the internal IP address of your CacheGuard appliance. A default system CA certificate is automatically generated the first time you start your CacheGuard appliance. It is recommended that you regenerate this CA certificate or import your own CA certificate and configure it as the system CA. You can use the following commands to generate a system CA certificate and its associated private RSA key:

- *tls ca system generate*
- *apply*

To import a CA certificate and its associated private RSA key into your CacheGuard appliance and set it as the system CA certificate, you must first place them on a trusted file server. A trusted file server is one that is authorised to exchange files with the CacheGuard appliance (refer to the access command for instructions on declaring a trusted file server). You can then load them into your CacheGuard appliance from that trusted file server. Assuming that your CA certificate and its associated private key are named *cg-ca.certificate* and *cg-ca.key* respectively, and are stored on a trusted SFTP file server with the IP address *172.18.2.1*, the following commands will import them into your CacheGuard appliance:

- *password file add sftp 172.18.2.1 john*
- *tls ca system load certificate sftp 172.18.2.1 cg-ca.certificate*
- *tls ca system load key sftp 172.18.2.1 cg-ca.key*
- *apply*

Exception Lists

To prevent MITM attacks, some HTTPS websites use a technique known as SSL pinning. SSL pinning embeds the HTTPS certificate directly into the web application, thereby preventing any MITM actions - and consequently

preventing SSL mediation.

Fortunately, CacheGuard appliance can be configured to bypass SSL mediation for certain predefined domain names (**deny** policy) or to operate exclusively on a predefined list of domain names (**allow** policy). You can create exceptions by directly specifying domain names (the quick method) or by using URL lists. To manage URL lists, refer to the urllist command. For example, to enable SSL mediation for the *example.com* domain name only, you may use the following commands:

- *sslmediate policy allow*
- *sslmediate exception urllist raz*
- *sslmediate exception domainname raz*
- *sslmediate exception domainname add example.com*
- *apply*

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Web Authentication

Web authentication can be used together with the **web** and **rweb** modes to restrict web browsing and/or access to reverse websites to authenticated users only. By enabling web authentication mode, explicit users of the web proxy in forwarding/browsing mode (**web** mode) and/or external users of reverse websites in reverse mode (**rweb** mode) are required to authenticate before being allowed to connect to the CacheGuard appliance. The CacheGuard appliance supports LDAP and Kerberos authentication modes, which you can configure using the authenticate command.

LDAP Authentication

In LDAP authentication mode, the first time a web user tries to access a web page, they are prompted to submit their credentials via a pop-up window opened by the web browser. With this authentication method, called basic authentication, the CacheGuard appliance sends the collected credentials to an LDAP server. If the credentials are validated by the LDAP server, the CacheGuard appliance allows the web user to connect. Otherwise, access is rejected. You can activate web authentication by using the mode authenticate on command.

To configure LDAP authentication, you must specify at least one LDAP server and the LDAP request that should be sent to it to authenticate users. In addition, as most LDAP servers require clients to bind before accepting requests, you must configure LDAP binding in your authentication settings. For example, the commands below configure LDAP authentication to send the '*ou=people,dc=example,dc=com*' '*uid*' '*userPassword*' '*objectClass=inetOrgPerson*' request to the LDAP server with hostname and IP address *ldap.example.com 10.0.10.201*. As this LDAP server requires binding, LDAP binding is activated by specifying the '*cn=admin,dc=example,dc=com*' bind DN (Distinguished Name).

- *mode authenticate on*
- *authenticate web on*
- *authenticate mode ldap on*
- *authenticate ldap request* '*ou=people,dc=example,dc=com*' '*uid*' '*userPassword*' '*objectClass=inetOrgPerson*'
- *authenticate ldap server add ldaps ldap.example.com 10.0.10.201*
- *authenticate ldap binddn* '*cn=admin,dc=example,dc=com*' *on*
- *apply*

LDAP Request Note: In the example above, the LDAP request specifies that users to retrieve are registered under the *ou=people,dc=example,dc=com* LDAP object, and the login name and password (provided by the web user during basic authentication) are compared against the values in the *uid* and *userPassword* LDAP attributes, respectively. Finally, users to search for must belong to the *inetOrgPerson* LDAP object class.

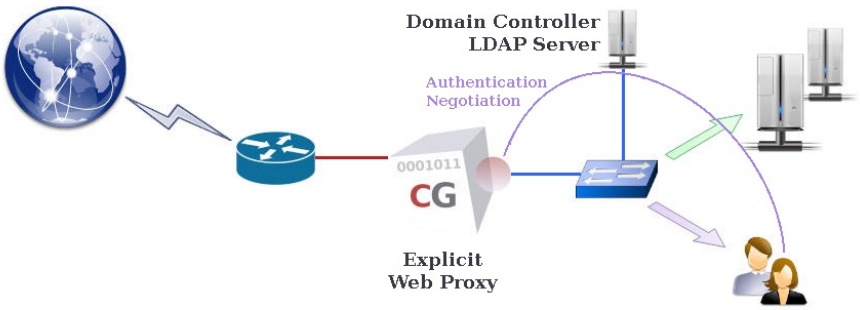
Note that if an empty value ('') is specified as the password LDAP attribute, an LDAP binding is performed during the basic authentication phase instead of comparing the entered password with the value stored in the password LDAP attribute (this is the preferred method used by Microsoft AD). If your LDAP server is Microsoft AD, you can use the following commands to configure LDAP authentication:

- *mode authenticate on*
- *authenticate mode ldap on*
- *authenticate ldap request* '*cn=users,dc=example,dc=com*' '*sAMAccountName*'
- *authenticate ldap server add ldaps ldap.example.com 10.0.10.201*
- *authenticate ldap binddn* '*cn=webgateway,cn=users,dc=example,dc=com*' *on*
- *apply*

Here, *cn=webgateway,cn=users,dc=example,dc=com* is the user used for LDAP binding. Make sure that the *cn=webgateway,cn=users,dc=example,dc=com* user exists in your Microsoft AD, is not disabled, and that its associated password never expires.

Kerberos Authentication

In Kerberos authentication mode, the credentials used by users to log in to their workstations are retrieved by the CacheGuard appliance during a negotiation phase and then used to authenticate web users. Kerberos-based authentication is a complex process that goes beyond this documentation, so we focus on how to configure the CacheGuard appliance to use Microsoft AD (as a Kerberos server).



You can use the following commands to configure Kerberos authentication on a CacheGuard appliance connected to a Microsoft AD server named *dc.example.com* on the network:

- *mode authenticate on*
- *authenticate mode ldap off*
- *authenticate mode kerberos on*
- *authenticate kerberos web proxy*
- *authenticate ad rdn 'cn=computers'*
- *domainname example.com*
- *authenticate kerberos server add dc.example.com*
- *authenticate kerberos encrypt aes*
- *apply*

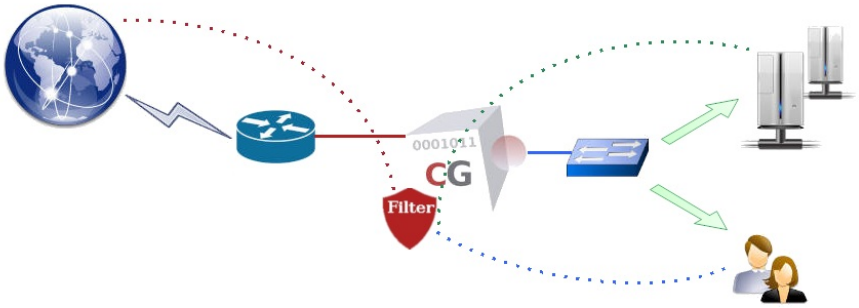
Web Proxy Specifications: In the example above, the authenticate ad rdn 'cn=computers' command sets the **rdn** (Relative Distinguished Name) of the object representing the *proxy.example.com* web proxy in the Microsoft AD directory tree, and the authenticate kerberos web proxy command sets the web proxy hostname to *proxy*. The web proxy's full name on the network would then be *proxy.example.com*, while *cn=proxy,cn=computers,dc=example,dc=com* represents its full DN in Microsoft AD.

It is **IMPORTANT** to note that the first time Kerberos authentication mode is activated (after the *apply* operation completes), it must be initialised on the CacheGuard appliance. During the initialisation process, if the LDAP object representing the web proxy in Microsoft AD does not yet exist, a new LDAP object for that web proxy is created in Microsoft AD. The initialisation process allows the CacheGuard appliance to obtain a Kerberos ticket. To initialise Kerberos authentication, use the authenticate kerberos create administrator command. This command requires interactive entry of the password associated with the used *administrator* account. The entered password is not permanently saved and is removed after obtaining the Kerberos ticket. Note that the Microsoft AD account used should have administrator permissions (for instance, the *administrator* user). Kerberos initialisation is an asynchronous operation executed in the background. The authenticate kerberos report command allows you to display a report on the Kerberos initialisation process.

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URL Guarding

URL guarding (or filtering) allows you to control web browsing within your organisation. You can activate URL guarding using the mode guard on command. This feature is based on blacklists (denied) and whitelists (allowed) of domain names and URLs that can be managed using the urllist command. You also have the option to use regular expressions in addition to URL lists. In this way, you can either allow access only to websites included in your whitelists or block access to URLs included in your blacklists. The guard command allows you to configure URL guarding by creating filters, policies, and rules.



A URL guard rule denies access to a set of blacklists or allows access to a set of whitelists according to filters associated with a policy. Each rule is associated with exactly one policy, identified by a unique name. A policy can be associated with one or more filters. A filter can specify a source IP address, an LDAP request, or a time specification. A policy named **default** allows you to define a default rule applied to all web clients for which no other policy matches. In the example below, a default rule is created to block access to Advertising and Malware websites for all web clients:

- *mode guard on*
- *urllist add Advertising*
- *urllist add Malware*
- *urllist load create Advertising tftp 172.18.2.1 Advertising domains urls*
- *urllist load create Malware tftp 172.18.2.1 Malware domains urls*
- *guard rule add default deny Advertising Malware*
- *apply*

In the example above, two URL list categories named *Advertising* and *Malware* are first created and then populated with lists of domain names and URLs stored on a remote TFTP file server with IP address *172.18.2.1*. Note that before being able to exchange files with a file server (TFTP, FTP, or SFTP), the file server must be declared as trusted. You can refer to the access file command to learn how to add a file server to the list of trusted servers. The sixth argument in the *urllist load...* command specifies the base name (*Advertising* or *Malware*) of files expected on the specified file server, and the urllist command expects to find files with the *.domains.gz* and *.urls.gz* extensions. As you can deduce, URL list files must be in gzip-compressed format (loading uncompressed URL list files is not supported).

Domain name files (*.domains.gz*) must contain a list of domain base names (one per line). A domain base name is a domain name without any prefix. For example, *example.com* is considered a domain base name, whereas *www.example.com* is not. To specify a domain name with a prefix, you should include it in a URL file (*.urls.gz*). URL files (*.urls.gz*) must contain a list of content specifications (one per line) similar to a URL without the protocol part, in the form <domain-name>/<path>, where <domain-name> is a fully qualified domain name and <path> is a path specification. For example, *www.example.com/foo/bar/zoo.html* is a valid URL in this context, while *http://www.example.com:81/foo/bar/zoo.html?id=1* is not.

URL Lists

URL guarding is primarily based on URL lists that should be kept up to date, as thousands of new websites appear on the web every day. Daily updated URL lists are offered as an optional service by CacheGuard Technologies, which you can easily subscribe to. However, you also have the option to use your own URL lists and keep them automatically updated on your CacheGuard appliance. To update a URL list, you can either recreate it entirely by loading its full content from a URL list file or simply load a difference file (preferred method). For example, to automatically update the URL list named *WebMail* from *ftp://172.18.2.1/DF/WebMail* on a daily basis, use the following commands:

- *password file add ftp 172.18.2.1 john*
- *urllist auto WebMail on load update daily ftp 172.18.2.1 DF/WebMail*
- *apply*

Difference URL list files must comply with the syntax described in the urllist command manual. Please refer to that manual for further information.

Policies & Filters

Guard policies allow you to apply URL guarding to specific groups of web clients matching certain criteria. For example, you may want to block Advertising websites for users in the *10.0.10.0 255.255.255.0* network who belong to the LDAP group *cn=support,ou=groups,dc=example,dc=com* between 12:30 and 13:30 (1:30 PM). To achieve this, you must first create three guard filters, then associate them with a policy, and finally create a guard rule based on that policy. The following commands define such a guarding policy:

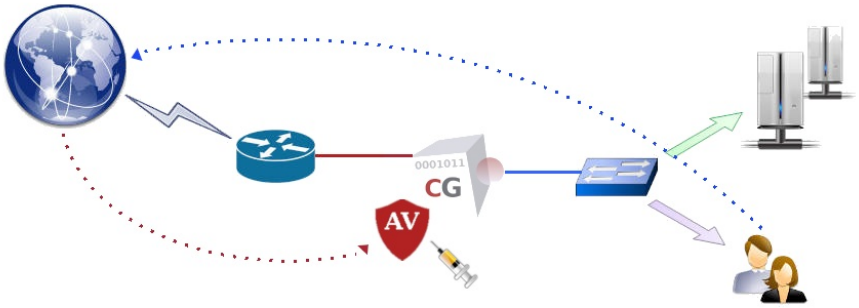
- *guard filter ip add london network 10.0.10.0 255.255.255.0*
- *guard filter time add lunchHours slot 12:30-13:30*
- *guard filter ldap add supportTeam 'cn=support,ou=groups,dc=example,dc=com' memberUid 'objectclass=posixGroup'*
- *guard policy add londonSupport ip london time lunchHours ldap supportTeam*
- *guard rule add londonSupport deny Advertising*
- *apply*

Note that LDAP filters can only be used when web authentication mode is activated. Refer to the Web Authentication section and the authenticate command manual for more information on LDAP authentication. **LDAP Filter Note**: in the example above, the *memberUid* parameter specifies the LDAP attribute used to store the login name provided by users during authentication. The *objectclass=posixGroup* defines the request used to retrieve the user information from the LDAP server.

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Antivirus

The antivirus detects malware (viruses, trojans, worms) in web traffic incoming from the **external** zone and blocks it at the gateway before it can enter your networks. The antivirus can operate in forwarding/browsing (**web**) mode as well as in reverse (**rweb**) mode. In forwarding mode, it blocks all attempts to access malware in web traffic, while in reverse mode, all attempts to upload malware to protected web servers are prevented (in **rweb** mode, the CacheGuard appliance is deployed in front of web servers). To activate the antivirus, use the mode antivirus on command followed by the apply command.



The antivirus detects MS Office macro viruses, mobile malware, and other threats. It supports 32/64-bit Portable Executable files and 32-bit ELF files. It scans not only simple files but also inspects content within archive and compression files such as, but not limited to, ZIP (+SFX), RAR (+SFX), TAR, GZIP, BZIP2, MS OLE2, MS Cabinet files (+SFX), MS CHM (Compiled HTML), MS SZDD compression format, BinHex, SIS (SymbianOS packages), AutoIt, and NSIS. In addition, the following file types are inspected:

- PE files compressed or obfuscated using tools such as Aspack (2.12), UPX (all versions), FSG (1.3, 1.31, 1.33, 2.0), Petite (2.x), PeSpin (1.1), NsPack, wwpack32 (1.20), MEW, Upack, and Y0da Cryptor (1.3).
- Almost every mail file format, including TNEF (winmail.dat) attachments.
- The most popular file formats such as MS Office and MacOffice files, RTF, PDF, and HTML.
- Various obfuscators, encoders, and files vulnerable to security risks, such as JPEG (exploit detection), RIFF (exploit detection), uuencode, and ScrEnc obfuscation.

Automatic Updating

The CacheGuard appliance periodically checks the malware signature database and, if necessary, downloads updates. Updates are retrieved from a public service named *database.clamav.net* on the internet. It is important to note that excessive or abusive download activity can result in temporary blocking by that service, which is outside the CacheGuard appliance's control. To avoid being banned, it is recommended to allow the CacheGuard appliance to automatically update the signature database and to avoid manual updates unless absolutely necessary.

To complement the standard malware signature database provided by *database.clamav.net*, additional malware signatures are available as an optional service from CacheGuard Technologies. You can easily subscribe to this service. After subscribing, simply activate it on your CacheGuard appliance by setting the provided password and file server name (using the access file and password file commands).

Antivirus and WAF

When the CacheGuard appliance is deployed as a WAF in front of your web servers (both **rweb** and **waf** modes activated), the antivirus scans all attempts to upload files to your protected or cloaked web servers. If malware is detected in an uploaded file, the CacheGuard appliance immediately blocks the upload before the file can reach the web servers. Note that the only upload method supported by the antivirus is the *POST* method with an encoding type of *multipart/form-data*. The following commands activate and configure the antivirus to scan file upload attempts on the web server with the IP address 10.20.0.100, protected by the CacheGuard appliance:

- *rweb site add www.example.com http*
- *rweb host www.example.com add rweb http 10.20.0.100*
- *mode rweb on*
- *mode waf on*
- *mode antivirus on*
- *apply*

Antivirus and MTA

The antivirus is natively used by the integrated web proxy to block malware in web traffic. However, it can also be used as a service offered to external clients or services such as an MTA (Mail Transfer Agent). For instance, to grant access to a remote exim4 MTA with the IP address 10.20.0.200 communicating with the CacheGuard appliance via its internal network interface, use the following commands:

- *ip internal 10.20.0.254 255.255.0.0*
- *port antivirus 8083*
- *access antivirus add internal 10.0.20.200 255.255.255.255*
- *apply*

In this example, the exim4 MTA must be configured to use the CacheGuard appliance as its antivirus service by adding the following line to its configuration file: *av_scanner = clamd: 10.20.0.254 8083*.

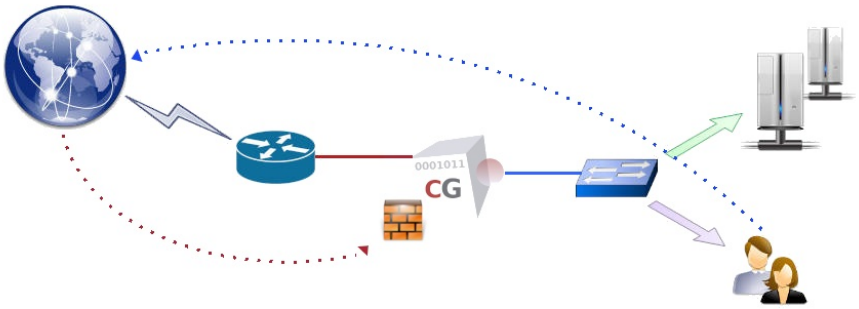
Testing the Antivirus

The European Expert Group for IT Security provides virus test files for verification purposes. These files are available on the https://www.eicar.org/ website. To test the antivirus using these files, download them to an HTTP (not HTTPS) web server and attempt to download them through your CacheGuard appliance web proxy. If your CacheGuard antivirus is properly configured, the download attempt should be blocked by the CacheGuard appliance. To test directly from the https://www.eicar.org/ website, which uses HTTPS (not HTTP), SSL mediation must be activated on your CacheGuard appliance. Please refer to the SSL Mediation section to learn how to enable SSL mediation.

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Network Security

CacheGuard appliance distinguishes between two types of network traffic: traffic exchanged with the appliance itself, and traffic that is only routed via the appliance (for which the source or destination is not the appliance itself). To control the traffic exchanged with the appliance, you can use the access command, while the firewall command can be used to control routed traffic. Both types of traffic are managed by a stateful firewall integrated into the appliance.



Administration Access

For security reasons, administration access (via SSH or the Web GUI) to a CacheGuard appliance is granted only to users or machines located within trusted networks. The same rule applies to file servers and monitoring systems that need to access the appliance. File servers are used to exchange files (via TFTP, FTP, or SSH) with the appliance for configuration or administration purposes (for instance, to load or save SSL certificates or perform a system backup). Monitoring managers can be used to monitor the appliance using SNMP.

In the example below, the appliance is configured to allow SSH and Web Administration GUI access via its **internal** network interface to remote users or machines in the *172.18.2.0 255.255.255.0* network. In addition, the SNMP manager with the IP address *172.18.2.202* is authorised to send SNMP requests to the appliance via its **internal** network interface, and the file server *ftp.cacheguard.net* is authorised to exchange files with the appliance via its **external** network interface.

- *access admin add internal 172.18.2.0 255.255.255.0*
- *access file add external ftp.cacheguard.net*
- *access mon add internal 172.18.2.202*
- *apply*

Note that to exchange files, CacheGuard appliance can use TFTP, FTP, or SFTP. As FTP and SFTP servers often require authentication (login and password), you can store authentication credentials in the appliance to avoid entering them interactively each time a file exchange is needed. Credentials may be defined using the access command or the password command. The following commands define credentials for FTP access to the *ftp.cacheguard.net* file server and SFTP access to the file server with IP address *172.18.2.203*:

- *password file add ftp ftp.cacheguard.net john*
- *access file add internal 172.18.2.203 sftp john*
- *apply*

Web Browsing Access

By default, the web proxy is accessible to all devices except those located in the external zone (routed via the **external** network interface). This default access policy can be modified using the access web command to restrict access to predefined networks only. To activate web proxy access limitation, you must define at least one web access rule. Note that when at least one previous peer or transparent network is defined, access limitation is implicitly activated, and you must explicitly specify permitted networks. Refer to the peer and transparent command manuals for more information about peers and transparent networks.

To restrict web proxy access to devices located in the *172.18.2.0 255.255.255.0* and *10.26.0.0 255.255.0.0* networks via the appliance's **internal** network interface, use the following commands:

- *access web raz*
- *access web add internal 172.18.2.0 255.255.255.0*
- *access web add internal 10.26.0.0 255.255.0.0*
- *apply*

The Firewall

CacheGuard appliance can act as a stateful firewall with NAT capabilities to filter routed network traffic according to source and destination IP addresses and protocols. To filter network traffic, you must define firewall rules. A firewall rule is attached to a network interface and controls incoming traffic through that interface. Rules attached to a network interface form a rule set, and there is one rule set per interface. To use the firewall, you must first activate it using the mode firewall on command and then define rules using the firewall command. In the absence of any rules, the following default rules are applied:

- New connections incoming from the **external** zone and destined for the **internal**, **auxiliary**, and **vpnipsec** zones are denied.
- New connections incoming from the **internal** zone (the **web** zone in VLAN mode) and destined for other zones are permitted.
- New connections incoming from the **auxiliary** zone are denied by default. To allow such traffic, you must explicitly define rules.
- In VLAN mode (mode vlan on), new connections incoming from the **rweb**, **antivirus**, **admin**, **mon**, **file**, and **peer** zones are denied by default. In non-VLAN mode (mode vlan off), these are all considered part of the **internal** zone and therefore follow its rules.
- New connections incoming from the **vpnipsec** zone and destined for the **internal** zone (the **web** zone in VLAN mode) are permitted. Connections incoming from the **vpnipsec** zone and destined for other zones are denied by default.

If at least one firewall rule is present in a rule set, default rules no longer apply, and only explicitly defined traffic will be allowed (or denied). Each firewall rule set implicitly includes a **deny any** rule at its end. For example, the following commands allow all TCP traffic incoming from the **web** interface with a source IP address in the 172.18.2.0/24 network and outgoing via the **external** network interface:

- *mode firewall on*
- *mode router on*
- *firewall web add allTCPToInternet allow tcp 172.18.2.0/24 external*
- *apply*

In the above example, if VLAN mode is activated, the defined rule applies exclusively to the **web** pseudo network interface (tagged VLAN). Otherwise, it applies to the native **internal** network interface.

When defining a firewall rule, you may also apply NAT (Network Address Translation) to the source and/or destination IP addresses of the traffic. For example, the following commands allow the IP address *192.168.44.55* to establish TCP connections via the **external** interface to the destination *192.168.22.11:80* and translate that destination to *10.0.10.11:81*:

- *mode firewall on*
- *mode router on*
- *firewall external add allWeb allow tcp 192.168.44.55 admin 192.168.22.11 80 nil 10.0.10.11 81*
- *apply*

It is **IMPORTANT** to note that in firewall rules, source and destination NAT are always applied **after** filtering.

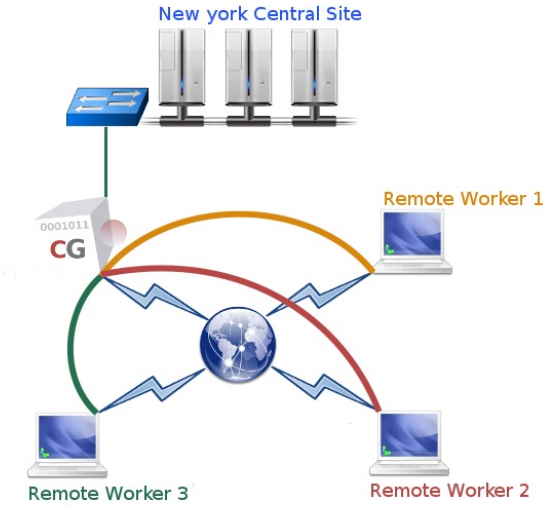
IPsec VPN

VPN stands for Virtual Private Network, and IPsec for internet Protocol Security. An IPsec VPN allows you to authenticate and encrypt data packets between private networks over a public IP network (for instance, the internet) to provide secure encrypted communication. You can create persistent IPsec VPNs between sites or enable remote workers to access your internal infrastructure securely via an IPsec VPN. CacheGuard appliance integrates an IPsec VPN server that you can activate using the mode vpnipsec on command. The vpnipsec command is then used to configure the IPsec VPN server.

Two types of IPsec VPNs are supported: site-to-site VPNs and remote-access VPNs. A site-to-site (or inter-site) VPN enables a permanent secure tunnel between two sites, allowing computers in both locations to communicate securely as if they were in the same place. To establish such a VPN tunnel, you require two VPN servers: a local VPN server and a remote (peer) VPN server.

A remote-access VPN enables remote users to connect securely to a central VPN server. Such tunnels allow remote workers to access systems protected by the VPN server securely, as if they were physically connected to the internal network.

Note that on a CacheGuard appliance, you must choose between site-to-site mode or remote-access mode; both cannot be activated simultaneously. The vpnipsec access on and vpnipsec access off commands enable you to switch between modes.



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WAN Optimisation

A CacheGuard appliance uses several technologies to optimise WAN (and LAN) bandwidth usage to improve availability and Quality of Service (QoS). Technologies such as HTTP compression and web caching help save network bandwidth during web browsing, while traffic shaping can reserve bandwidth for critical applications. All these features can be activated simultaneously on the same CacheGuard appliance.

Web Caching

Web caching is one of the core and historical features of a CacheGuard appliance. It stores browsed (or served, in reverse mode) web content on disks before sending it to web clients. Cached web content can then be retrieved locally instead of being downloaded again from the web, significantly saving network bandwidth. You can use the mode cache on command to activate web caching.

Cached Object Size

For performance reasons, a CacheGuard appliance caches only objects within a defined size range. The cache command can be used to modify the upper and lower cache size limits.

It is important to adjust these limits with care: caching all objects (even very small ones) may decrease performance, as disk I/O can become slower than network I/O. Similarly, caching overly large objects can quickly saturate the cache with content rarely reused.

To configure web caching to store only objects between 20 KB and 51200 KB, use the following commands:

- *cache object 20 51200*
- *apply*

Caching Big Objects

In some cases, it is useful to cache very large objects that are shared by many users, such as operating system updates for PCs or smartphones. To cache large objects (greater than 256 MB - the default upper limit), you must enable the big-object caching feature and define its limits.

Big-object caching reserves a specific area of disk space for large files, preventing them from occupying the entire cache. The size of this area depends on your disk capacity and installation parameters.

To enable caching for objects between 131073 KB (128 MB) and 2097152 KB (2 GB), use:

- *cache bigobject on 131073 2097152*
- *apply*

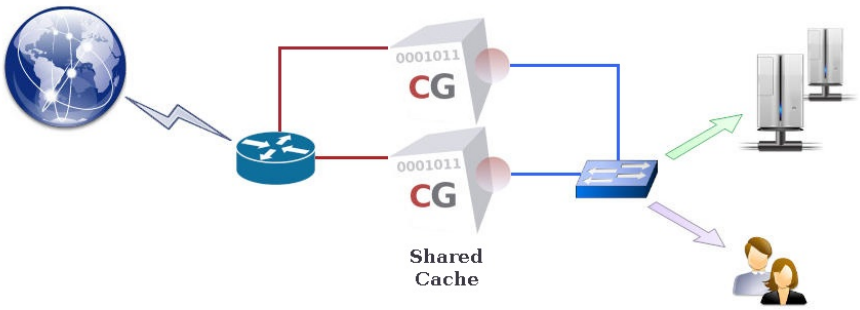
Peer Caches

Two or more CacheGuard appliances can be paired to share their respective web caches. Use the peer command to configure peering between appliances. A paired CacheGuard appliance is referred to as a **peer**.

Peers can be configured in parallel or chained modes:

- **Parallel mode:** The appliance checks its own cache first. If the requested content is not found, it queries its peers before fetching it from the internet.
- **Chained mode:** The appliance checks its own cache first. If the object is not found, it always queries its chained peers (called next peers) instead of the internet directly. Each remote peer must be configured to accept connections from its previous peers.

If more than one next peer is defined, the appliance distributes requests among them in a load-balanced manner.



HTTP Compression

A CacheGuard appliance can compress web traffic to reduce its size and save bandwidth. HTTP compression is particularly beneficial for low-bandwidth networks prone to packet loss, such as Wi-Fi or satellite-based links.

When HTTP compression is enabled, the appliance compresses textual content (HTML, JavaScript, XML, etc.) before sending it to clients. Modern browsers automatically decompress this content.

To enable HTTP compression, use the mode compress on command.

Traffic Shaping

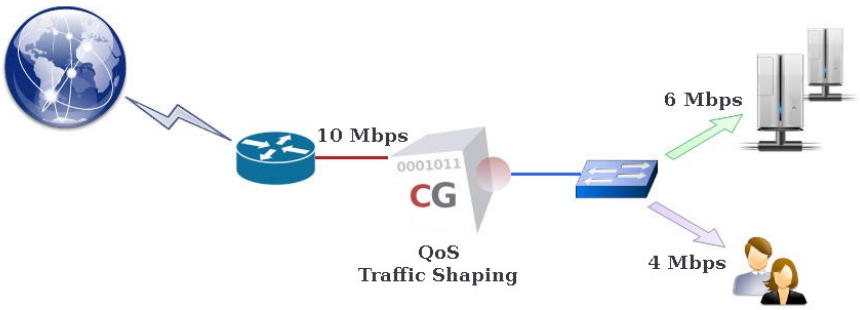
A CacheGuard appliance includes a bandwidth manager that allows you to shape and schedule network traffic. The system distinguishes between:

- Traffic exchanged directly with the appliance.
- Traffic routed through the appliance (where the source or destination is external).

Use the mode qos on command to enable traffic shaping, and configure it using the qos command.

Traffic types that can be managed include:

- **antivirus**: Traffic between the embedded antivirus and external services.
- **file**: File transfers between the appliance and file servers.
- **peer**: Traffic between CacheGuard peers.
- **rweb**: Reverse proxy traffic between the appliance and real web servers.
- **tweb**: Transparent web proxy traffic between the appliance and web clients.
- **web**: Forward proxy traffic between the appliance and web clients.
- **vpnipsec**: Encrypted IPsec VPN traffic.
- **default**: Any other traffic not matched by the above categories.



To configure bandwidth limits, first define the total available (incoming and outgoing) bandwidth for each interface in kbps, then assign percentages or absolute values to each traffic type.

For example, to set the total bandwidth of the external interface to 10,000 kbps (10 Mbps) and other interfaces to 1,000,000 kbps (1 Gbps):

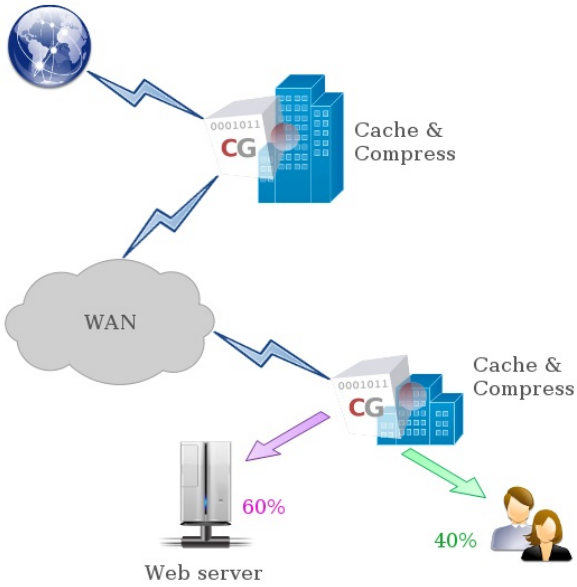
- *mode qos on*
- *qos bandwidth external ingress 10000*
- *qos bandwidth external egress 10000*
- *qos bandwidth internal ingress 1000000*
- *qos bandwidth internal egress 1000000*
- *qos bandwidth auxiliary ingress 1000000*
- *qos bandwidth auxiliary egress 1000000*
- *apply*

To reserve 40% of external bandwidth for web browsing and 60% for reverse web traffic:

- *qos shape web external ingress 40%*

- *qos shape web external egress 40%*
- *qos shape rweb external ingress 60%*
- *qos shape rweb external egress 60%*
- *apply*

Bandwidth Borrowing



Traffic shaping can be strict or flexible. In flexible mode, unused reserved bandwidth for one traffic type can be temporarily borrowed by others.

For example, if 60% of external bandwidth (6 Mbps) is allocated to reverse web traffic but only 4 Mbps is in use, the remaining 2 Mbps can be borrowed by other traffic types.

To enforce strict allocation (no borrowing), disable it as follows:

- *qos borrow external ingress off*
- *qos borrow external egress off*
- *apply*

QoS Fine-Tuning

Traffic shaping can be fine-tuned to prioritise specific networks. For instance, you can reserve 90% of browsing bandwidth for one network and limit another to 10%.

This setting can be defined contextually, such as when using the access web command.

Example:

- *qos shape web internal ingress 30%*
- *qos shape web internal egress 30%*
- *access web add internal 10.26.0.0 255.255.0.0 90%*
- *access web add internal 172.18.2.0 255.255.255.0 10%*
- *apply*

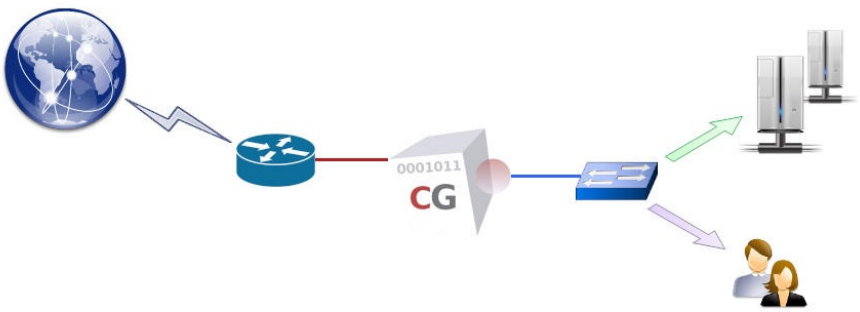
You can combine web caching, peering, HTTP compression, and traffic shaping to achieve maximum bandwidth optimisation.

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Web Server Cloaking

A CacheGuard appliance can be implemented as a reverse web proxy positioned in front of a group of web servers to protect them against unwanted direct access on the one hand, and to optimise the web traffic exchanged with them on the other. This implementation is referred to as *reverse mode* (or **rweb** mode), and the protected web applications are known as *reverse websites*. All security and optimisation features available in CacheGuard-OS can be combined to provide a highly secure and efficient network optimisation solution for protecting web applications. The following list summarises the advantages of this mode when combined with others:

- **Firewall**: blocks unauthorised network access to web servers.
- **Reverse Proxy**: prevents direct exposure of web servers to the internet.
- **SSL Offloading**: terminates SSL connections on the CacheGuard appliance to relieve web servers of encryption tasks.
- **WAF**: the web Application Firewall (WAF) blocks content-based attacks such as SQL injection attempts.
- **Load Balancing**: distributes website traffic evenly across multiple web servers.
- **High Availability**: ensures service continuity in the event of a failure on a CacheGuard appliance or a web server.
- **Caching**: stores static web objects such as images to reduce load on web servers.
- **HTTP Compression**: conserves WAN bandwidth.
- **Antivirus**: prevents malware injections into web pages.
- **QoS**: enables traffic shaping and provides Quality of Service to critical applications.



To activate the reverse mode, use the mode rweb on command. You must then use the rweb command to configure this mode. For example, the following commands activate reverse mode and protect the web server with the IP address *10.0.10.11* hosting the *www.example.com* website:

- *mode rweb on*
- *ip external 192.168.1.1 255.255.255.0*
- *rweb site add www.example.com http*
- *rweb host www.example.com add rweb http 10.0.10.11*
- *apply*

Anti-Spoofing Note: a host is always associated with a specific network interface through which it is authorised to communicate, thereby preventing IP address spoofing. The **rweb** keyword in the command "*rweb host www.example.com add **rweb** http 10.0.10.11*" identifies that network interface. It corresponds to the pseudo-interface named **rweb** in VLAN mode, or the native **internal** interface if VLAN mode is disabled. For detailed information, refer to the rweb command manual.

In reverse mode, the CacheGuard appliance acts as a virtual web server, concealing the real web servers (referred to as hosts). These cloaked web servers must then be publicly accessible via the CacheGuard appliance's external IP address. For example, in the configuration above, the *www.example.com* name should publicly resolve to 192.168.1.1.

Load Balancing

If multiple hosts are associated with a single website, the total load is distributed among all associated hosts. The CacheGuard appliance continuously monitors host availability, and in the event of a host failure, it temporarily removes that host from the load-balancing pool, thereby maintaining high availability for the website.

The default load-balancing method is round-robin, which distributes an equal number of requests to each host. When associating a host with a website, an optional weight may be specified to adjust the load distribution. Furthermore, if the underlying web application requires that a specific web client always be served by the same host (to preserve application context), sticky load balancing must be enabled for that website. Sticky connections are based on either an existing HTTP cookie or one inserted by the CacheGuard appliance, which can be configured using the rweb

command.

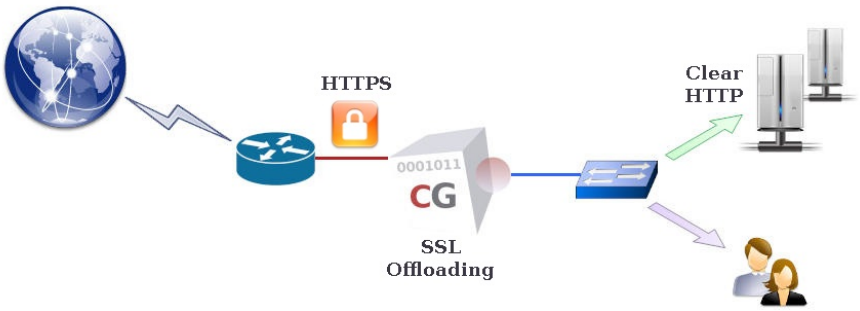
To extend the previous example, the following configuration adds a host with the IP address *10.0.10.12* to the pool of load-balanced hosts and enables round-robin sticky load balancing for the *www.example.com* website:

- *rweb host www.example.com add rweb http 10.0.10.12*
- *rweb balancer www.example.com robin sticky*
- *apply*

SSL Offloading

To allow a CacheGuard appliance to process (inspect, cache, etc.) web content within HTTPS (encrypted) traffic, it must first decrypt that traffic to gain unencrypted access to the data. To achieve this, the CacheGuard appliance acts as an SSL terminator for the HTTPS websites it manages. While communication between web clients and HTTPS websites managed by the appliance is always encrypted, communication between the appliance and hosts may be either encrypted (using HTTPS) or unencrypted (using HTTP).

When the HTTP protocol is used for communication with hosts, the CacheGuard appliance performs SSL offloading for those hosts. It is crucial in such configurations to ensure that the unencrypted web traffic between the appliance and hosts is fully isolated, preventing any unauthorised access. The **http** keyword in the command "*rweb host www.example.com add rweb* ***http*** *10.0.10.12*" specifies the use of SSL offloading for that host. To avoid SSL offloading, replace **http** with **https**. Refer to the rweb command manual for further details.



As an SSL offloader, the CacheGuard appliance must have access to the public SSL certificates and private keys associated with the HTTPS websites it manages. The tls command can be used to manage the SSL certificates and private keys used by the appliance. A website can exist in both HTTP and HTTPS versions. In such cases, all HTTP requests are redirected to the HTTPS version by default (although this behaviour can be modified if required). Continuing the previous example, the following commands add SSL encryption capability to the *www.example.com* website (accessible at *https://www.example.com/*):

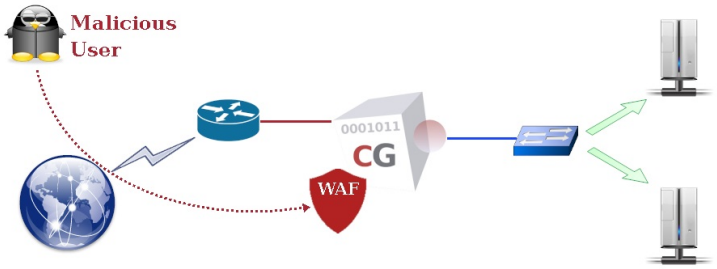
- *tls server add myTLSObject*
- *tls server generate myTLSObject*
- *rweb site add www.example.com https myTLSObject*
- *apply*

TLS Note: once the *apply* operation is complete, an RSA private key, a self-signed X.509 SSL certificate, and a CSR (Certificate Signing Request) are generated. The CSR can be saved to a trusted file server and submitted to a Certificate Authority (CA) for signing. The new signed certificate can then be imported into the CacheGuard appliance. Refer to the access file and tls server command manuals for guidance on saving and loading SSL certificates.

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The WAF

WAF stands for Web Application Firewall and protects web applications from malicious web requests intended to compromise or damage applications and/or servers. CacheGuard appliance integrates a WAF that works jointly with the Web Server Cloaking to provide an enhanced level of security for web servers. The WAF can protect against content attacks such as, but not limited to, XSS (Cross-site Scripting), SQL injection and command injection. Malicious requests are blocked before they reach real web servers (hosts).



To enable the WAF, both **rweb** and **waf** modes must be activated using the mode rweb on and mode waf on commands. You can refer to the Web Server Cloaking section to learn how to configure the **rweb** mode. Then use the waf command to configure content filtering for protected web applications. The CacheGuard WAF provides two types of content filtering that may be activated concurrently on the same appliance. The two content-filtering types are as follows:

- **Generic Filtering**: rejects known content attacks; it may be activated globally (for all websites) or per protected website.
- **Customised Filtering**: allows you to permit or deny web requests that you design yourself on a per-website basis.

Generic Filters

Generic WAF rules are organised in groups called filters (a filter is composed of WAF rules). Each blocking rule in a generic filter has a score (an integer) and whenever a rule matches in web traffic its score is added to a global tally. Once a threshold score is reached for a given transaction, the WAF blocks the traffic. Generic filters are provided by OWASP and are classified as follows:

- **dliis**: rules to prevent data leakage from IIS applications.
- **dljava**: rules to prevent data leakage from Java applications.
- **dlphp**: rules to prevent data leakage from PHP applications.
- **dlsql**: rules to prevent data leakage from SQL requests.
- **java**: rules to protect Java applications.
- **lfi**: rules to protect against Local File Inclusion attacks.
- **nodejs**: rules to protect Node.js applications.
- **php**: rules to protect PHP applications.
- **rce**: rules to protect against Remote Code Execution attacks.
- **rfi**: rules to protect against Remote File Inclusion attacks.
- **sf**: rules to protect against Session Fixation attacks.
- **sqli**: rules to protect against SQL injection attacks.
- **xss**: rules to protect against cross-site scripting attacks.

Generic filters may be activated or deactivated globally. For example, to activate the WAF and protect all websites (cloaked by the reverse web proxy in **rweb** mode) against SQL injection attacks, use the following commands:

- *mode rweb on*
- *mode waf on*
- *waf generic sqli on*
- *apply*

If no generic WAF filter is specified for a website, that website is protected by globally activated generic WAF filters. If a particular website requires specific protection, you may activate or deactivate generic WAF filters for that website. For example, to deactivate the **sqli** generic filter for the *www.example.com* website, use:

- *waf rweb generic www.example.com sqli off*
- *apply*

Custom Filters

Custom WAF filters provide precise control over web requests for a specific website. A custom WAF filter consists of custom WAF rules defined in a textual file that you load into a CacheGuard appliance. A custom WAF rule permits or denies a web request according to its HTTP method and content. A rule is defined in one, two or three lines depending on the specified HTTP method.

A custom WAF rule must begin with the **rule** keyword followed by an identifier, an action (**allow** or **deny**) and an HTTP method (in lowercase). Supported HTTP methods are *GET*, *HEAD* and *POST*. For the *GET* and *POST* methods, a second optional line may specify allowed contents (the path and arguments) in the web request. That second line must begin with the **uri** keyword and be followed by a PCRE (Perl Compatible Regular Expression) that specifies allowed contents. For the *POST* method, a third optional line may specify allowed contents in the *POST* body. That line must begin with the **body** keyword and be followed by a regular expression specifying allowed data in the body of the *POST* request.

As an example, the following custom WAF filter includes six rules: the first and second rules allow *GET* requests on "*/*" and "*/index.html*" respectively. The third rule allows *POST* requests to "*/cgi-bin/set-phone.cgi*" with a body matching "*name=<string>&phone=<numbers>*". The final three rules deny any other web requests.

```
rule r1 allow get
uri "^/$"

rule r2 allow get
uri "^/index\.html$"

rule r3 allow post
uri "^/cgi-bin/set-phone.cgi$"
body "^name=[[:print:]]*\&phone=[[:digit:]]*$"
```

```
rule r4 deny get
rule r5 deny head
rule r6 deny post
```

To apply this custom WAF filter to the *www.example.com* website, save it on a trusted file server and then load it into your CacheGuard appliance and associate it with the *www.example.com* website. Refer to the access file command manual to learn how to declare a trusted file server. The following commands load a custom WAF filter from a file named *www.example.com.rules* on a TFTP server at *172.18.2.1* and apply it to the website:

- *waf rweb custom www.example.com load tftp 172.18.2.1 www.example.com.rules*
- *apply*

When a custom WAF filter is loaded, the custom WAF rule compiler validates its content and rejects the load if an error is detected. Note that when generic WAF filters are combined with a custom WAF filter, generic filters are applied first. Malicious requests are therefore rejected by generic filters before reaching the custom filter.

Reputation Filters

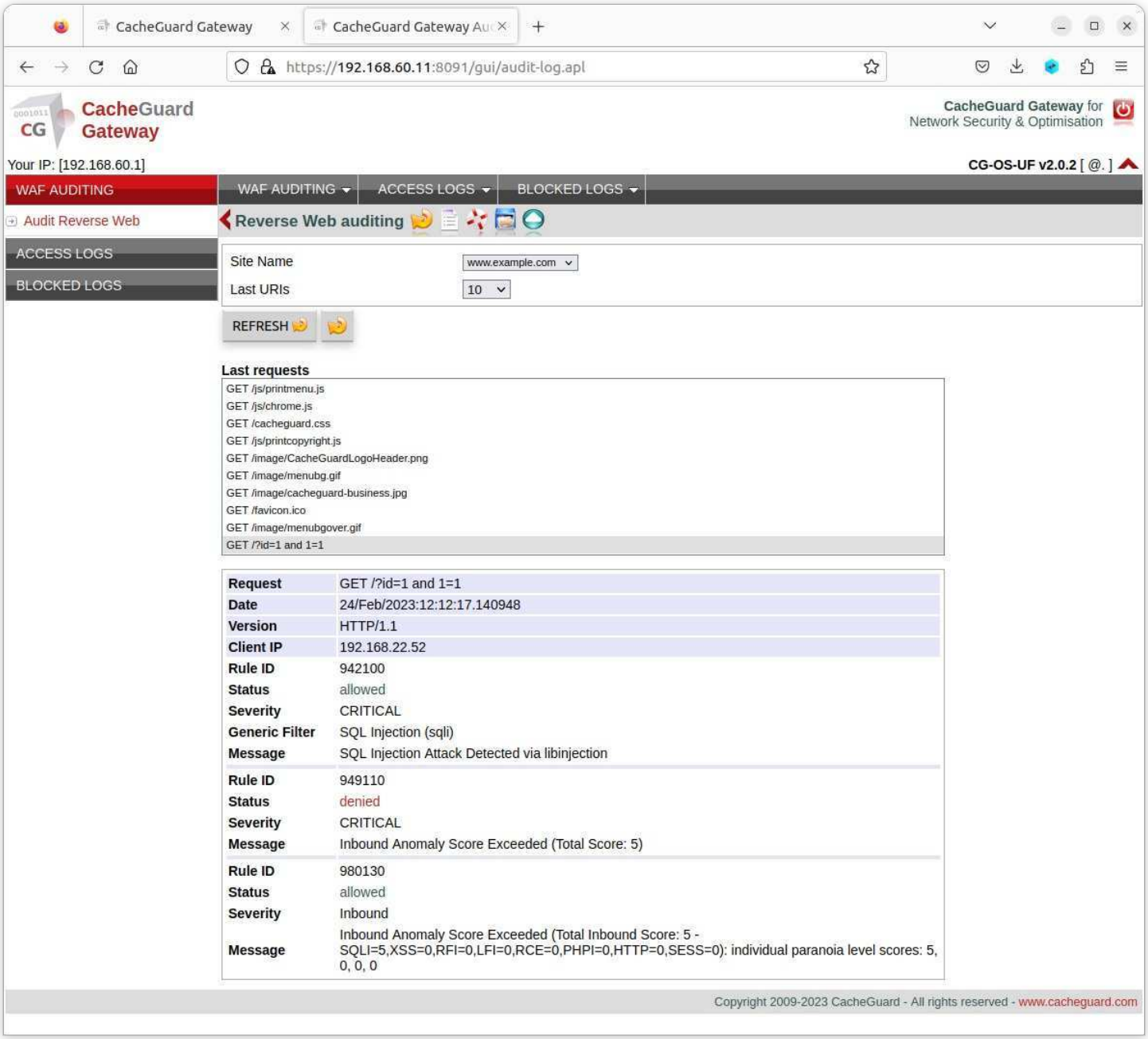
CacheGuard appliance can block web requests originating from IP addresses with a poor reputation. Reputation-based filtering can block requests from specific countries or from entries listed in an RBL (Real Time Blacklist). See the waf command manual for more information on reputation-based filtering.

Website Auditing

A Web Auditing GUI is integrated into the CacheGuard appliance and permits inspection of HTTP requests for a given website. Web auditing helps you determine why a web request was blocked and which WAF rule (generic or custom) caused the block. To enable web auditing for the *www.example.com* website, use the following commands:

- *admin waudit on*
- *waf rweb audit www.example.com on*
- *apply*

The Web Auditing GUI is accessible via *https://<cacheguard-ip-address>:8091* where *<cacheguard-ip-address>* is the CacheGuard appliance IP address. The IP address to use depends on the administration access policy and topology configured on the appliance. Refer to the access admin and admin topology command manuals for further information. The screenshot below shows the Web Auditing GUI.

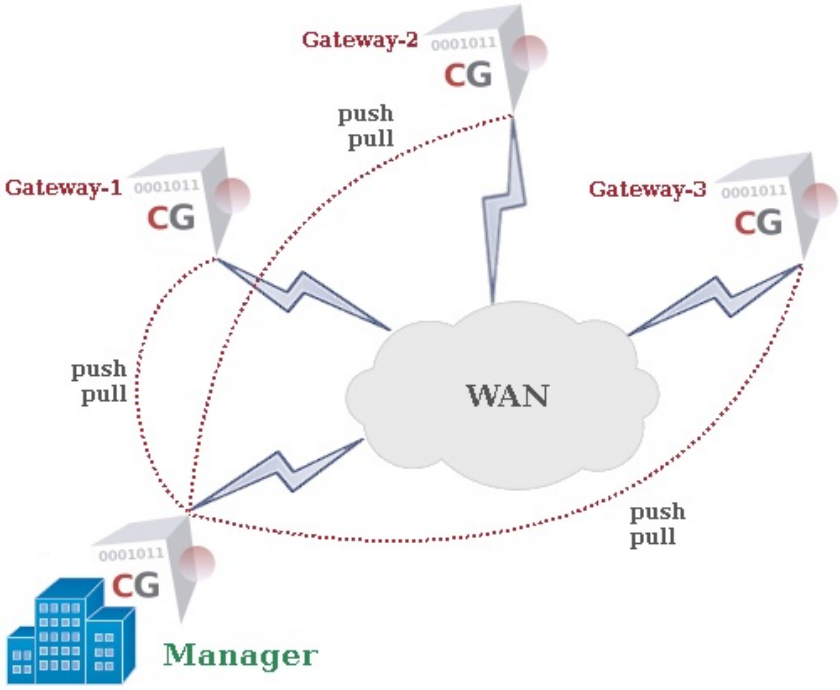


Sometimes a WAF rule unintentionally matches legitimate traffic (a false-positive). Web auditing assists in detecting such false positives so they may be addressed either by modifying the web application (when feasible) or by bypassing the rules that cause the false positives. Refer to the waf command manual to learn more about WAF rule bypassing.

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Using a Manager

If you deploy several CacheGuard Gateways in your organisation, you have the option to configure them separately, one by one. However, in most cases, Gateways deployed within the same organisation share similar configurations, requiring you to repeat the same configuration process for each device. A CacheGuard Manager system enables you to centrally configure and manage multiple remote Gateways from a single point. With a Manager, you can create configuration templates and build Gateway configurations based on those templates. These configurations can then be pushed in parallel to several Gateways with just a few commands or clicks.



If you also need to automatically update data such as URL lists, the Manager allows you to download them once and push them simultaneously to all managed Gateways. The Manager provides a uniform and optimised method to manage your Gateways efficiently.

Gateway Access

Before being able to manage Gateways from a Manager, the Gateways must permit management access from that Manager. The Manager uses the SSH protocol and SSH keys to connect to Gateways. Therefore, Gateways must allow the Manager's IP address SSH access and authorise the Manager's SSH public key (so that a password is not required for each SSH session). A Gateway can be managed by one master Manager and, optionally, by a backup Manager. The backup Manager always maintains a hot copy of all Gateway configurations stored on the master Manager and can be activated in the event of a failure on the master Manager.

A Manager system has only one logical network interface, called **internal**. To obtain the Manager's IP address and its SSH public key, use the following commands on the Manager:

- *ip internal*
- *manager ssh show*

To allow a master Manager with the IP address 192.168.1.22 and the SSH public key 'ssh-rsa AAAAB3Nza...' to manage a Gateway via its external network interface, use the following commands on the Gateway:

- *manager add master external 192.168.1.22 'ssh-rsa AAAAB3Nza...'*
- *apply*

Gateway Enrolment

Once a Manager is authorised to access a Gateway, the first step is to enrol that Gateway on the Manager and, optionally, pull its current configuration to store it on the Manager. Gateway configurations on the Manager are identified by unique identifiers that must be defined during enrolment. In addition, Gateways are organised into groups known as domains, and each Gateway must belong to exactly one domain. This enables you to push or pull configurations in parallel across all Gateways within the same domain.

To enrol and pull the configuration of a Gateway with the IP address 10.0.10.254, use the following commands on the Manager:

- *manager gateway add my-company gateway-1 10.0.10.254*
- *manager gateway pull my-company gateway-1*

In this example, my-company and gateway-1 are the domain name and identifier respectively for the enrolled Gateway. The pull operation runs in the background. To obtain a report on the latest pull operation, use the following command:

- *manager gateway report pull*

Gateway Configuration

Gateway configurations can be modified on the Manager and then pushed to remote Gateways. To begin editing a Gateway configuration on the Manager, use the following command:

- *manager gateway begin conf gateway-1*

This command takes you inside the Gateway configuration context, where you can use the same commands normally available on a Gateway system. Once configuration is complete, use the **apply** command to verify and validate the configuration's integrity, and then use the **end** command to return to the Manager configuration level. At this stage, you can push the new configuration to the remote Gateway with the following command:

- *manager gateway push my-company gateway-1*

The push operation runs in the background. To obtain a report on the latest push operation, use the following command:

- *manager gateway report push*

Working with Templates

The main strength of the Manager lies in its ability to work with templates. A template is a special Gateway configuration that can be applied to multiple Gateways. This allows you to quickly configure several Gateways with similar configurations, customising only the elements that differ between them (such as IP addresses). To create a template named my-template on a Manager system and begin configuring it, use the following commands:

- *manager template add my-template*
- *manager template begin conf my-template*

Once the template configuration is complete, use the **apply** command to verify and validate its integrity, and then use the **end** command to return to the Manager configuration level.

Now that you have created a template, you can apply it to a Gateway configuration. To do this, you must be within a Gateway configuration context. To configure a managed Gateway using a template named my-template, use the following command:

- *conf manager template my-template*

Master & Slave Managers

To ensure high availability and service continuity, a slave Manager can be configured to maintain a hot copy of all data stored on the master Manager. In this way, if the master Manager fails, the slave Manager can be activated to continue managing the Gateways. To enable communication between the master and slave Managers, both must know each other's IP address. Additionally, the slave Manager must be aware of the master Manager's SSH public key to allow SSH connections. Assuming the master Manager has the IP address 192.168.1.22 and the slave Manager has 192.168.1.33, use the following commands on the master Manager:

- *manager sync role master*
- *manager sync peer 192.168.1.33*
- *apply*

Then, use the following commands on the slave Manager:

- *manager sync role slave*
- *manager sync peer 192.168.1.22 'ssh-rsa AAAAB3Nza...'*
- *apply*

Here, 'ssh-rsa AAAAB3Nza...' represents the SSH public key of the master Manager.

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